Soil and Water Testing Methods

A Laboratory Manual

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FOREWORD

Soil testing in India had its modest beginning in 1955-56 with the setting up of 24 soil testing laboratories under the overall supervision and control of the Indian Agricultural Research Institute, New Delhi, valuable material support as well as technical assistance being provided by the United States of America through its Technical Co-operation Mission (later on USAID). As a result of the growing interest, the programme made rapid strides and today the country is having **one of the world's largest chain of soil testing laboratories,** both static and mobile, operated by Central and State Governments, Agricultural Universities, Research Institutions, Fertilizer Industry as well as some cooperative and private organisations, with the IARI having continued to take the responsibility of technical coordination.

Realising the great importance of the programme and the need for trained personnel, the IARI, in collaboration with the USAID (TCM) initiated, as early as in 1958, a **Short Course Training in Soil Testing** with a view to have a uniform standard in the laboratory procedures and also to provide a wider perspective in the concepts of soil testing to the scientists working in the various laboratories. Since then the Division of Soil Science and Agricultural Chemistry of the Institute has been regularly organising the Refresher Training Course for the benefit of the Soil Chemists incharge of the soil testing laboratories. The **pioneering role** played by Dr. S.P. Raychaudhuri, the then Head of the Division in initiating the programme and by (Late) Dr. R. V. Tamhane, Dr. N. P. Datta and Dr. B. Ramamoorthy, all former Special Officer (Soil Testing) and later on Head of the Division and also by Dr. Gilbert R. Muhr of the USAID in the development of soil testing work in India and in the organisation of the Training Course has been unique and will be remembered by all those who are engaged in research and extension of soil testing service in the country. Dr. T.D. Biswas, former Head of the Division, was mainly responsible for the reorientation and strengthening of the course curricula particularly with regard to instrumentation, which is an important component of soil test methodology. So far, **more than 400 scientists** from all parts of the country have undergone

this training in **twenty-five** such courses conducted at this Institute; the **XXV Course** has just been completed early this month.

It is considered very appropriate to mark this great occasion of **25 years of the training programme** by publishing a suitable **Laboratory Manual on Soil and Water Testing Methods**. From what once used to be a few cyclostyled sheets of instructions to the trainees, the lecture materials and practical exercises have been gradually enlarged in size and revised over the years in keeping with the expansion in the scope of the training. The group of scientists led by Dr. A.B. Ghosh who have been primarily responsible for the efficient conduct of the Training Course for the last several years has spared no efforts in updating this publication and making it quite comprehensive and purposeful for the benefit of the scientists, technical staff and students engaged in soil testing work.

It is hoped that this Manual will serve the need to fulfil a self-contained laboratory guidebook in soil test methodology and meet the long-felt demand.

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Grateful thanks are expressed to Dr. N.N. Goswami, Head of the Division of Soil Science and Agricultural Chemistry, IARI for giving his valuable suggestions in revising and updating the Manual to its present form and constant all round support. The foreword kindly written by him visibly reflects his keen interest in its publication on the occasion of **Twenty-five years of Short Course Training in Soil Testing** in this Institute.

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COLLECTION AND PREPARATION OF SOIL SAMPLES FOR ANALYSIS

The importance of having a truly representative sample can be very well realized from the fact that only a minute fraction of the huge soil mass of the field is actually used for analysis in the laboratory. While collecting samples the following aspects should be carefully considered.

The sample must properly represent the area. A field can be treated as a single sampling unit only if it is appreciably uniform in all respects. Variation in slope, colour, texture, crop growth and management should be taken into account and separate sets of composite samples need to be collected from each of such area. Recently fertilized plots, bunds, channels, marshy tracts and spots near trees, wells, compost piles or other non-representative locations must be carefully avoided during sampling.

The root penetration of plants is an important consideration for ascertaining the depth of sampling. Fields growing cereals and other seasonal crops are normally sampled up to 0-15 cm (generally expressed as plough layer). For deep rooted crops like sugarcane, horticultural plants or under dry farming conditions it may be necessary to obtain samples from different depths or layers.

In case of saline-alkali soils, salt crusts (visible or suspected) on the soil surface should be sampled separately and the depth of sampling recorded. Generally the sample may be drawn to the plough depth.

Sampling Procedure

To obtain a composite sample, small portions of soil are to be collected up to the desired depth (0-15 cm or more) by means of suitable sampling tools from at least 10 to 15 well distributed spots (from each sampling unit) after scrapping off the surface litter, if any. When crops are planted in rows samples can be drawn in between the lines, if not recently fertilized.

The soil collected in this manner should be thoroughly mixed (by hand) on a clean piece of cloth, polythene sheet or thick paper and the bulk reduced by quartering and about 500 g (1/2 kg) of the composite sample is retained. The soil must be quickly air dried in shade at room temperature and put in cloth or polythene bags with suitable description and identification marks.

Sampling tools

Samples can be drawn with any of these hand implements namely, (i) soil tube (auger), (ii) screw type auger, (iii) post-hole auger, (iv) *kassi* (spade) and (v) *khurpi* (a common gardening hand tool).

For sampling of soft and moist soil the tube auger, spade or *khurpi* is quite satisfactory. A screw type auger may prove more convenient on hard/dry soil while the post-hole auger is useful for sampling excessively wet area like rice field. If a spade or *khurpi* is used, a V-shaped cut may be first made (up to the plough layer) and a uniformly 1.5 cm (approx.) thick slice is taken out. Augers are necessary for sampling from lower depths.

The various steps involved in the proper collection of representative soil samples in the field and their handling are depicted in figure 1.

Precautions in collection and storing of samples : Care in handling the soil samples against contamination is extremely important. Special precautions are necessary if the samples are to be tested for micronutrient elements. Any possibility of contact with chemicals, fertilizers or manures must be avoided. Cotton, jute or plastic bags which had previously been containing fertilizers, salt or lime should not be used at any stage. Soil samples should preferably be stored in clean cloth or polythene bags. Glass, porcelain or polythene jars are useful in keeping few selected samples for long duration. The bags can be re-used only after thorough washing and drying.

Information sheet : In addition to the location, field number, name of cultivator and identification marks, relevant information regarding slope, drainage, irrigation, previous cropping history, fertilizer use etc. of the field (s) must be recorded and sent alongwith the soil samples. This is a specific requirement in soil testing work and must be insisted upon. A specimen of the information sheet to be filled up is given in appendix. I. This can be partly modified if felt necessary.

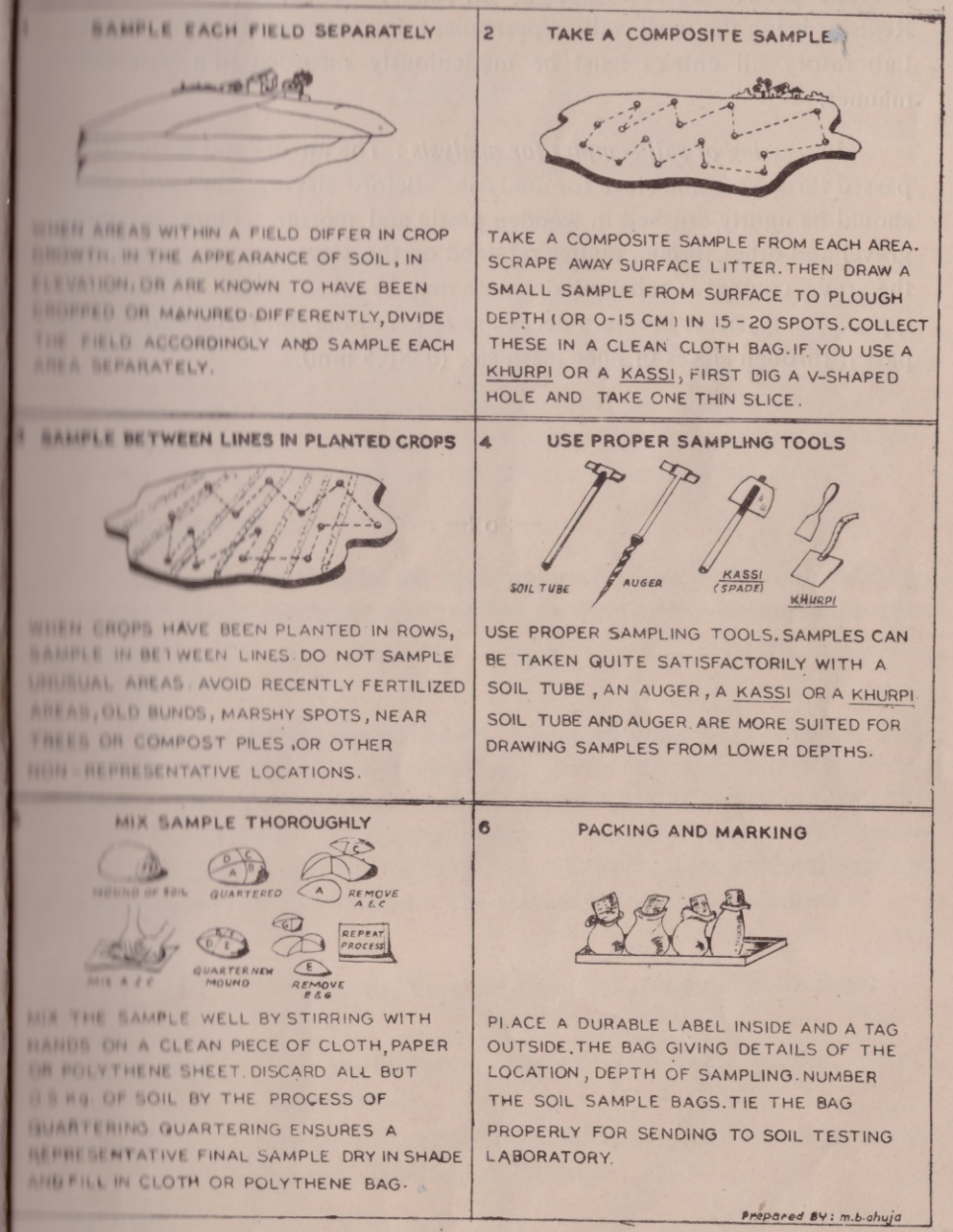


Fig. 1. Procedure for collection and handling of soil samples

Registration of samples : As soon as the samples arrive at the laboratory these are to be serially registered giving the date and other relevant particulars furnished in the information sheet. The Sample Register being the most valued permanent document of the Soil Testing Laboratory all entries must be meticulously recorded in a systematic manner.

Processing of soil samples for analysis : The air-dry soil is normally passed through 2 mm sieve for analysis. Before sieving, the soil clods should be lightly crushed in wooden pestle and mortar. Plant residues, gravel and other foreign matter retained on the sieve are discarded. If the gravel content is substantial a note may be taken of that. For certain types of analysis, it becomes necessary to grind the soil further so as to pass it through sieves of finer mesh size (0.2-0.5 mm).

—: o :—

Type of soil is India

1. Red soil
2. Clay
3. Alluvium
4. Entisol

Type

1. Alluvial soil
2. Black soil
3. Red soil
4. Laterite soil
5. Arid soil
6. Forest & mountain soil
7. Marsh soil

Section A

SOIL TEST METHODOLOGY

Some selected physical, physico-chemical and chemical tests such as determination of texture, pH (soil reaction), electrical conductivity, mineralizable nitrogen and available phosphorus and potassium are usually carried out to evaluate such of the soil properties which are known to have a direct impact on the productivity of the soil as well as its fertility status. A qualitative assessment of the free calcium carbonate content, if any, is also made in case of neutral and alkaline soils.

I. SOIL TEXTURE

The texture is determined by a rapid procedure by feel or rubbing the soil between the thumb and the fingers. Proficiency is gained through experience, trial and error and comparison with samples of known textural class.

Steps for texture determination : A small quantity of the dry soil is moistened and mixed thoroughly on a glass or porcelain dish so as to form a soft ball and then worked until stiff and squeezed out between thumb and fore-finger. The feel to fingers, ease of forming ball, stickiness or grittiness, whether forming soil ribbons or merely crumbling on squeezing etc. are to be noted. The texture is classified as follows :

Very light

Sand : Very gritty, does not form ball, does not stain finger.

Light

Loamy sand : Very gritty, forms ball but very easily broken, stains finger slightly.

Sandy loam : Moderately gritty, forms fairly firm ball which is easily broken, definitely stains finger.

Medium

Loam : Neither very gritty nor very smooth, forms firm ball but does not ribbon, stains finger appreciably.

Silt loam : Smooth or sticky, buttery feel, forms firm ball, stains and has a slight tendency to ribbon with flaky surface.

Heavy

Clay loam : Moderately sticky, slightly gritty feel, forms moderately hard ball when dry, stains, ribbons out on squeezing but the ribbon breaks easily.

Silty clay loam : Same as above but very smooth, shows flaking on ribbon surface, similar to silt loam.

Very heavy

Clay : Very sticky feel, forms ball which when dry can not be crushed by fingers, stains heavily, squeezes out at right moisture into long (2-5 cm) ribbon.

The textural class of known soil samples taken for comparison is correctly ascertained from the percentage distribution of sand, silt and clay determined according to the international pipette method (Piper 1966) and using the standard triangular diagram.

II. DETERMINATION OF SOIL REACTION (pH)

The pH value which is a measure of the hydrogen (or hydroxyl) ion activity of the soil-water system indicates whether the soil is acidic, neutral or alkaline in reaction. Since crop growth suffers much both under very low (strongly acidic) as well as high (alkali) pH, suitable reclamation measures become necessary. Presence of neutral soluble salts (as in saline soils) is not normally reflected in the pH but when in large excess they tend to suppress the ionic activity.

Instrument

The instrument for pH measurement commonly used is a glass electrode pH meter with calomel reference electrode including salt bridge. Most digital pH meters have single (combined) electrode assembly. The instrument being a potentiometer, the pH scale has to be calibrated before use with buffer solutions of known pH values.

Since the dissociation/ionic activity increases with temperature the pH scale of 0 to 14 is valid only for a particular temperature (25°C). Thus all instruments are provided with temperature setting/correction knob which must be made use of.

Reagent

Standard buffer solutions : These may be of pH 4.0, 7.0 or 9 2 in pure water or in other ranges of expected pH value. In case of buffer tablets (available commercially) a single piece is to be dissolved in freshly double distilled water and made up to 100ml. It is necessary to prepare a fresh buffer after every few days as the solutions do not keep for long. A 0.05M analytical grade (AR) potassium hydrogen phthalate ($KHC_8H_4O_4$ mol. wt. 204.14) gives a pH of 4.0 (actually 3.97) at 25°C and this can also be used as a standard buffer. A stock solution of 0.25M is prepared by dissolving 51.04 g of the pure dry salt in warm water and making up to 1 litre. Three to four drops of toluene are added to prevent growth of mould. A dilution of 10 ml of this solution to 50 ml with water gives 0 05M solution.

Procedure for pH measurement

(*a*) *pH in saturated soil paste* : This is prepared by adding small amounts of distilled water to a portion of soil while working with a spatula. At saturation, the soil paste glistens, flows slightly when the container is tilted, slides freely and cleanly off the spatula. After mixing, it should be allowed to stand with a cover above the container for about an hour. Free water must not collect on the soil surface nor should the paste stiffen markedly or lose its glistening appearance on standing. The pH meter is to be set for the temperature and calibrated with two buffers, one in the acidic side and the other in alkaline or neutral range. The electrodes (glass and calomel or combined) are carefully inserted in the paste and pH measurement is made. The electrode is moved a little to ensure removal of water film around it and pH reading is again taken when it is constant.

(*b*) *Soil to water ratio of 1:2* : 20g of soil is taken in a 100 ml beaker to which 40 ml of distilled water is added. The suspension is stirred at regular intervals for 30 minutes and the pH is recorded. The suspension must be stirred well just before the electrodes are immersed and readings taken.

(*c*) *pH in 0.01M $CaCl_2$ solution* : To 20g of soil in a beaker 40 ml of 0.01M $CaCl_2$ solution (1.470g AR $CaCl_2.2H_2O$/litre) is added. The suspension is shaken thoroughly (30 minutes) and the pH is recorded. The reading is shifted downwards in comparison to the pH value obtained in water suspension. With acid soils it normally ranges from 4 to 5 and

in calcareous ones between 6.9 and 7.2. The pH of near neutral soil in $CaCl_2$ solution is usually around 6.

N.B. Before each determination the electrodes must be washed with a jet of distilled water and water removed from the surface with a piece of filter or blotting paper.

III. MEASUREMENT OF ELECTRICAL CONDUCTIVITY

Since ions are the carrier of electricity, the electrical conductivity (EC) of the soil-water system rises according to the content of soluble salts in the soil giving rise to more of ion pairs on dissociation as it happens in case of a dilute solution. Thus the measurement of EC can be directly related to the soluble salts concentration of the soil at any particular temperature.

Instrument

The apparatus for measuring electrical conductivity consists of an AC salt bridge or electrical resistance bridge and conductivity cell having electrodes coated with platinum black. The instrument is also available as an already calibrated assembly (solubridge) for giving directly the conductivity of solutions in millimhos per cm at 25°C.

Reagent

Standard potassium chloride solution: 0.7456g of dry reagent grade potassium chloride is dissolved in freshly prepared double distilled water and made to one litre. At 25°C it gives an electrical conductivity of 1411.8×10^{-6} (0.0014118) mhos/cm or 1.41 mmhos/cm. The conductivity bridge is to be calibrated and cell constant determined with the help of this solution. Even if the scale is marked directly in mmhos/cm (as in solubridge) it is necessary to check and calibrate the instrument with the KCl solution.

Procedure

20g of soil is shaken intermittently with 40 ml of distilled water in a 150 ml Erlenmeyer (conical) flask for one hour and allowed to stand. Alternatively, the clear extract after pH determination can be used for electrical conductivity measurement. The conductivity of the supernatant liquid is determined with the help of the salt (conductivity) bridge. The measurement of EC (expressed in mmhos/cm) is to be adjusted for a known temperature (usually 25°C) of the solution by setting the knob provided for this purpose.

Test for calcium carbonate

A qualitative rapid test for free calcium carbonate ($CaCO_3$) is made by observing the intensity of effervescence (due to CO_2 evolution) when about 10ml of dilute hydrochloric acid (1+3) is added to a small quantity (about 5g) of air dry soil taken on a watch glass. The amount of $CaCO_3$ present in the soil can be broadly categorized as (i) traces, (ii) low, (iii) medium (iv) high and (v) very high on the basis of the effervescence.

For soils showing high and very high degree of effervescence a separate quantitative assay can be made by reacting a known weight of sample with a known volume of excess standard ($N/_2$ or $N/_5$) HCl and back titration (after warming and shaking) of unreacted acid with standard alkali (Piper 1966). The nett quantity of acid consumed can be attributed to the presence of free calcium carbonate.

IV. BIOLOGICAL AND CHEMICAL METHODS FOR ESTIMATION OF AVAILABLE NITROGEN

The major part (more than 90%) of soil nitrogen exists in complex combination in the organic matter (humus) fraction. It becomes available to crops after breakdown to simple forms followed by mineralization. This type of transformation is mostly biological in nature. Therefore, methods involving the determination of mineral forms of nitrogen (NH_4-N and NO_3-N) produced on incubation of soil under aerobic and anaerobic conditions are generally followed. These are considered to be quite satisfactory as index of nitrogen availability in arable soils. Incubation under waterlogged conditions may be preferred over aerobic incubation. Besides this, estimation of CO_2 produced or evaluation of the microbial growth or pigment production on incubation of soil samples with nitrogen-free readily decomposable organic material or nutrient media and microbial cultures, is also carried out.

The chemical methods for determination of soil nitrogen availability involve either direct or indirect estimation of total N or measurement of a fraction of easily hydrolysable nitrogen by employing solutions of dilute acids or alkali. Distillation with alkaline potassium permanganate solution has often been adopted for estimating the readily oxidisable and reactive forms of soil nitrogen.

One of the most widely used rapid soil testing techniques for the assessment of available nitrogen is based upon the estimation of readily oxidisable organic carbon which roughly represents 58% of the soil

organic matter, the seat of nitrogen in soil. This approach has been found to work fairly well unless the bulk of the organic matter is non-humic in nature and the organic carbon values are on the very high side.

1 Nitrogen mineralization (ammonification and nitrification) under aerobic conditions (Keeney and Bremner 1966)

Procedure : 10g of air-dry soil (sieved through 2 mm) is intimately mixed with 30g of acid-washed quartz sand (30-60 mesh) and transferred to a 250 ml wide-mouth square type bottle containing 6 ml of distilled water and mixed thoroughly. The material is distributed evenly on the bottom. The bottle is closed with a rubber stopper having a central hole through which a glass/plastic tube is connected to the mixture (soil and sand) for aeration. The incubation is carried out at 30°C for 14 days. At the end of this period, the stopper is removed and 100 ml of 2N KCl is added and the neck of the bottle is fitted back with a solid rubber stopper. The suspension is shaken for one hour on a mechanical shaker and filtered (Whatman No.1). 20 ml aliquot of the clear filtrate is taken in a 250 ml boiling flask to which about 80 ml of distilled water plus 0.2g each of MgO (freshly ignited and cooled) and Devarda's alloy are added and distilled at a moderate speed. The distillate is collected in a 150 ml conical flask containing 20 ml of 2% boric acid solution having 20 ml/l methyl red-bromocresol green mixed indicator (as given under 3 on page 11 in case of alkaline $KMnO_4$-N) and the ammonia liberated titrated against standard acid. The amount of $(NH_4+NO_3+NO_2)$ nitrogen present in the soil without incubation is deducted from the figure for incubated soil to obtain the quantity of N released. A blank determination without soil is also necessary.

2. Anaerobic incubation (Subbiah and Bajaj 1962; Warring and Bremner 1964)

(Ammonia released under waterlogged conditions)

Procedure for micro estimation : 12.5 ± 1.0 ml of distilled water is placed in a 16 mm × 150 mm test tube or 150 ml conical flask and 5g of air-dry (2mm) soil is added to this. The tube/flask is stoppered and incubated at 40°C for seven days. The contents are transferred to a conical flask (150 ml) and 25 ml of 4N KCl or 50 ml of 2N KCl is added and shaken (1/2 hr) mechanically. The suspension is filtered immediately.

A suitable aliquot of the clear filtrate is distilled with 0.5g of MgO (freshly ignited) and the distillate collected in a conical flask containing 20 ml of boric acid along with mixed indicator (as in the case of aerobic incubation). The distillate is titrated against standard acid (0.02N H_2SO_4). The ammonia content of the soil without incubation is determined according to the same procedure and deducted from the amount obtained after incubation.

Macro estimation : For this purpose, 20g of soil is mixed with equal weight of quartz sand in 100 ml conical flask. The soil is submerged with distilled water to a level of 2 cm above the surface. It is incubated for one week at 35°C. Thereafter the contents are leached with 200ml of 1N NaCl and ammonia (displaced) is estimated by distilling with freshly ignited MgO.

The procedure has been found suitable as an index of nitrogen availability for rice, *bajra* (pearl millet) and wheat and also as a single extractant for the simultaneous extraction of NPK for rice.

3. Mineralizable nitrogen (Alkaline permanganate method) (Subbiah and Asija 1956)

The procedure involves distilling the soil with alkaline potassium permanganate solution and determination of the ammonia liberated which serves as an index of the available/mineralizable nitrogen status.

Reagents required

(i) 0.32% potassium permanganate solution
(ii) 2.5% sodium hydroxide
(iii) Liquid paraffin (extra pure)
(iv) 0.02N sulphuric acid
(v) 2% boric acid solution containing 20 ml of mixed indicator per litre
(vi) Mixed indicator : 0.066g methyl red plus 0.099g bromocresol green dissolved in 100 ml of 95% alcohol

Procedure

In a 800 ml dry Kjeldahl flask, 20g of soil is taken. To this 20 ml of water is added followed by 100 ml each of 0.32% $KMnO_4$ and 2.5% NaOH solutions. The frothing during boiling is prevented by liquid

paraffin (1 ml) and bumping by adding a few glass beads. The contents are distilled in a Kjeldahl assembly at a steady rate and the liberated ammonia collected in a conical flask (250 ml) containing 20 ml of boric acid solution (with mixed indicator). With the absorption of ammonia the pinkish colour turns to green. Nearly 100 ml of distillate is to be collected in about 30 minutes which is titrated with 0.02N H_2SO_4 to the original shade (pinkish). Blank correction (without soil) is to be made for the final calculation.

Calculation

$$\text{Mineralizable N (kg/ha)} = R \times 0.02 \times \frac{1}{20} \times 0.014 \times 2.24 \times 10^6 = R \times 31.36$$

where R = volume of 0.02N H_2SO_4 required for titration

V. ESTIMATION OF ORGANIC CARBON

1. Walkley and Black (1934) rapid titration

The organic matter (humus) in the soil gets oxidized by chromic acid (potassium dichromate plus conc. sulphuric acid) utilizing the heat of dilution of sulphuric acid. The unreacted dichromate is determined by back titration with ferrous (ammonium) sulphate (redox titration).

Reagents

(i) *1N potassium dichromate* (49.04g of AR grade $K_2Cr_2O_7$ per litre of solution)

(ii) *0.5N (approx.) ferrous ammonium sulphate* (196g of the hydrated crystalline salt per litre containing 20 ml of conc. H_2SO_4). This solution is relatively more stable and convenient to work than that of ferrous sulphate.

(iii) *Diphenylamine indicator* : 0.5g diphenylamine dissolved in a mixture of 20 ml of water and 100 ml of conc. H_2SO_4

(iv) Concentrated sulphuric acid (sp.gr. 1.84) containing 1.25 per cent silver sulphate (In case of soils free from chlorides use of Ag_2SO_4 can be avoided)

(v) Ortho-phosphoric acid (85%) and/or sodium fluoride (chemically pure).

Procedure

The soil is ground and completely passed through 0.2 mm sieve and 1.00 g is placed at the bottom of a dry 500ml conical flask (Corning/ Pyrex). 10 ml of 1N $K_2Cr_2O_7$ is pipetted in and swirled a little. The flask is kept on asbestos sheet. Then 20 ml of H_2SO_4 (containing 1.25% Ag_2SO_4)is run in and swirled again two or three times. The flask is allowed to stand for 30 minutes and thereafter 200ml of distilled water is added. After incorporation of 10 ml of phosphoric acid or 0.5g sodium fluoride and 1ml of diphenylamine indicator the contents are titrated with ferrous ammonium sulphate solution till the colour flashes from blue-violet to green. A combination of H_3PO_4 and NaF is found to give a sharper end point. Simultaneously, a blank is run without soil. If more than 7 ml of the dichromate solution is consumed the determination must be repeated with a smaller quantity (0.25-0.50g) of soil.

Calculation

$$\text{Organic carbon (\%)} = \frac{10(B-T)}{B} \times 0.003 \times \frac{100}{\text{wt. of soil}}$$

where B = volume (in ml) of ferrous ammonium sulphate solution required for blank titration; and

T = volume of ferrous ammonium sulphate needed for soil sample.

2. Colorimetric method of estimation (Datta *et al.* 1962)

The oxidation of soil organic matter is carried out by dichromate-sulphuric acid mixture and the intensity of the green colour of the chromium sulphate formed is measured to give directly the amount of carbon oxidised.

Instrument

Photo-electric colorimeter (double cell) preferably of Klett-Summerson type

Reagents

(i) *1N (AR grade) potassium dichromate* (49.04 g/1)

(ii) Concentrated sulphuric acid (sp. gr. 1.84) with 1.25 g of silver sulphate per 100 ml (not needed if chlorides are absent)

(iii) Sucrose (AR grade), anhydrous

Procedure

One gram of soil (passed through 0.2 mm sieve) is taken in a dry 100 ml conical flask (Pyrex/Corning). 10 ml of 1N $K_2Cr_2O_7$ is added and swirled a little followed by 20 ml of conc. sulphuric acid and swirled again. After keeping for 30 minutes on an asbestos sheet the content in the flask is carefully centrifuged to a clear state. The green chromium sulphate colour of the supernatant layer is read in the colorimeter after adjusting the blank solution (without soil) to zero using 660 mμ (red) filter.

Calculation

Organic carbon (%)

= Colorimeter reading (R) × factor (from standard curve)

Preparation of standard calibration curve

One to 25 mg of anhydrous sucrose (AR) are weighed out and placed in separate 100 ml dry conical flasks. 10 ml of 1N $K_2Cr_2O_7$ solution is added to each followed by 20 ml of conc. H_2SO_4. The flasks are swirled a little and kept for half an hour. Thereafter readings are taken in the colorimeter using 660 mμ filter after adjusting the blank to zero. A curve is drawn plotting the concentrations of carbon (in sucrose) and colorimeter readings (R) and the factor is calculated from sucrose. With Klett-Summerson colorimeter the factor comes to 0.0042 with possibility of little variation.

Note : Instead of centrifuging (after addition of $K_2Cr_2O_7$ and H_2SO_4 to soil) the flasks can be kept overnight and the supernatant solution carefully decanted off for colorimeter reading. This avoids the chance of any damage to the centrifuge due to accidental spilling of the concentrated acid and a large number of samples can be handled at the same time. If chromic acid crystals appear, specially in cold climate, the addition of 10ml of water to each flask after half an hour's reaction has been found to be helpful before keeping them overnight. The standard curve must be prepared in an identical manner. However, for clayey soils centrifuging becomes necessary as these do not settle properly for colorimetric measurements.

VI. DETERMINATION OF NITRATE NITROGEN

(Phenol disulphonic acid method)

A nitrate in contact with sulphuric acid produces nitric acid which in dry condition (in presence of excess concentrated sulphuric acid) causes nitration of phenol disulphonic acid. The product is like a

nitrophenolic type indicator giving intense yellow colour in alkaline medium which can be subjected to colorimetric determination. The reaction must proceed *in cold* otherwise nitric acid may be lost by volatilization.

Instrument

Photo-electric colorimeter (double cell type)

Reagents

(i) Calcium sulphate ($CaSO_4 \cdot 2H_2O$) or calcium oxide (chemically pure)

(ii) *Phenol disulphonic acid* (*in sulphuric acid*) : This is prepared by taking 25g of pure phenol (C_6H_5OH, crystal white) in a dry conical flask (500 ml) or tall beaker (Corning/Pyrex) to which 150 ml concentrated AR sulphuric acid (nitrate-free) and 75 ml AR fuming sulphuric acid (nitrate-free) are mixed very carefully and the contents are kept on boiling water bath for 2 hours, covered with a watch glass. On cooling, the phenol disulphonic acid (in excess sulphuric acid) is stored in amber coloured bottle. In case fuming sulphuric acid is not available the same amount of concentrated sulphuric acid may be added instead and kept on the boiling water bath for at least six hours.

(iii) Liquor ammonia (laboratory reagent 0.88 sp. gr., diluted with equal volume of water)

(iv) Potassium nitrate (analytical grade)

Procedure

20g of soil is shaken continuously with 50 ml distilled water for one hour in a 100 ml conical flask fitted with rubber stopper. A pinch of $CaSO_4$ or CaO is added and again shaken thoroughly for a few minutes to help in quick settling of the soil, and the contents filtered through a dry filter paper (Whatman No. 1 or equivalent). First few ml of the filtrate may be rejected. 20 ml of the clear aliquot is transferred to a 50 ml porcelain or silica dish, evaporated to dryness on steam bath and cooled to room temperature. Three ml of phenol disulphonic acid reagent is allowed to react with the residue by rotating the dish. After 10 minutes, 15 ml of distilled water is added and stirred with a glass rod.

On cooling, the contents are washed down into 100 ml volumetric flask or stoppered cylinder. Ammonia (1 : 1) is to be added slowly with mixing till the solution is alkaline as indicated by the development of yellow colour due to the presence of nitrate. Then another 2 ml of ammonia is added and the volume made up (100 ml) with distilled water. Intensity of yellow colour is read in the photo—electric colorimeter using 420 mμ (blue) filter.

Preparation of standard curve for nitrate : A stock solution containing 100 ppm nitrate-nitrogen (NO_3-N) is prepared by dissolving 0.7215 g of AR grade potassium nitrate (oven-dried and cooled) and made to one litre. This is diluted ten times to give a 10 ppm NO_3-N solution. Aliquots (2, 5, 10, 15, 20 and 25 ml) are evaporated on boiling water bath to dryness in small porcelain dishes (or beakers). When cool, 3 ml of phenol disulphonic acid is added and the yellow colour is developed and read as described above. A blank (without nitrate) must be run and correction made by adjusting the colorimeter to zero with blank. A calibration curve is drawn between concentration of NO_3-N and colorimeter reading.

Calculation

$$NO_3\text{—N (kg/ha)} = \mu g \text{ of } NO_3\text{—N in test solution} \times \frac{50}{20} \times \frac{1}{20} \times \frac{2.24 \times 10^6}{10^6}$$

$$= \mu g \text{ of } NO_3\text{—N in test solution} \times 0.28$$

VII. ESTIMATION OF AVAILABLE PHOSPHORUS (IN SOIL)

Phosphorus in soil occurs as orthophosphate in several forms and combinations and only a small fraction of the total amount present may be available to plants which is of direct relevance in assessing the phosphorus fertility level.

A wide variety of chemical soil tests for available P has been proposed which extract variable quantities of phosphorus. These include, water, carbonated water, solution of neutral salts namely KCl and $CaCl_2$, strong mineral acids of different concentrations (2.5% HCl, 0.2N HNO_3, 0.2N H_2SO_4), weak organic acids like citric (1%), lactic and acetic (2.5%), buffered dilute acid solutions both organic and inorganic, such as Truog's reagent (0.002N sulphuric acid and ammonium sulphate pH 3.0), Egner's lactate extractant (ammonium lactate-lactic acid pH 3.8), Morgan's solution (sodium acetate-acetic acid pH 4.8) and alkaline

extractants such as NaOH (0.1N), K_2CO_3 (1 and 2%) and $NaHCO_3$ (0.5M). Use of chelating agents (like EDTA or DTPA) and specifiic chemicals for anionic exchange has been suggested for extracting certion forms of adsorbed and chemically bound phosphate in the soil system.

The extractants have been developed either from the concept of simulating the action of plant roots or to obtain a suitable measure of the readily soluble inorganic forms to represent the plant available soil phosphorus. Thus, the estimation of specific categories of soil P considered to be of significance to crop growth has lead to the introduction of neutral NH_4F (ammonium fluoride) solution as an extractant for adsorbed phosphate and NH_4F— HCl combination for adsorbed plus acid soluble form. A 0.5M $NaHCO_3$ solution adjusted to pH 8.5 has been designed to control the ionic activity of calcium through the solubility product of $CaCO_3$ in case of neutral and calcareous soils. In this process, the most reactive (high specific surface) form of P is extracted from the phosphates of iron, aluminium and calcium present in the soil.

In soil testing work, the available phosphorus content is readily determined by extraction with suitable reagents according to specified soil to solution ratio and time of shaking. In the filtered extract, phosphorus is estimated colorimetrically by adding ammonium molybdate and thereafter reducing the molybdenum-phosphate complex in acidic medium. The intensity of blue colour on reduction provides a measure for the concentration of P in the test solution. The two extractants and procedures normally used are given below.

1. **Olsen's method** (Olsen *et al.* 1954)

The extracting reagent for Olsen's P is 0.5 molar sodium bicarbonate (pH 8.5) prepared by dissolving 42.0g of $NaHCO_3$ (laboratory reagent) in distilled water to give one litre of the solution. The pH is adjusted to 8.5 with small quantities of 10% NaOH. The method has been found widely applicable in slightly acid, neutral and calcareous soils.

To 2.5g of soil in 100 ml conical flask a little of Darco G 60 or equivalent grade of activated carbon (free of phosphorus) is added followed by 50 ml of Olsen's reagent. A blank is to be run without soil. The flasks are shaken for 30 minutes on a platform type shaker and the contents filtered immediately through dry filter paper (Whatman No.1) into clean and dry beakers or vials. In the filtrate phosphorus is

estimated colorimetrically by Dickman and Bray's (excess acid) procedure.

Note : The activated carbon (even if marked phosphorus-free) is likely to contain traces of P which have to be removed by repeated washings with Olsen's reagent followed by warm distilled water. The material should test free of phosphorus when extracted with Olsen's reagent. The sodium bicarbonate should also be tested free from any phosphate contamination.

2. Bray's No. 1 method (Bray and Kurtz 1945)

This procedure is primarily meant for soils which are moderately to strong acid (pH around 5.5 or less). The Bray's No. 1 reagent consists of 0.03N NH_4F in 0.025N HCl solution. 2.775 g of NH_4F (AR/LR) is added to 2.5 litres of 0.025N HCl (checked against standard NaOH solution) and dissolved.

5g of soil and 50 ml of the reagent are shaken in 100 ml conical flask for exactly 5 minutes and filtered. To avoid interference of fluoride, 7.5 ml of 0.8M boric acid (50g of H_3BO_3 per litre) can be added to 5 ml of the extract, if necessary. In the extract, phosphorus is estimated colorimetrically by Dickman and Bray's method.

Colorimetric estimation of phosphorus (Dickman and Bray 1940)

Instrument

Photo-electric colorimeter (double cell, preferably of Klett-Summerson type)

Reagents

(*i*) *Dickman and Bray's reagent* (*for Bray's No. 1 method*) : 15g of ammonium molybdate (AR) is dissolved in 300 ml of distilled water, warmed to about 60°C and filtered, if necessary, after cooling. To this, 350 ml of 10N HCl is added and made up to one litre. The normality of the HCl should be adjusted correctly by titration.

(*ii*) *Dickman and Bray's reagent having excess of acid for Olsen's method* : 15g of ammonium molybdate (AR) is dissolved in 300 ml of warm water (about 60°C), cooled and filtered if necessary. To this, 400 ml of 10N HCl is added and made up to one litre.

(*iii*) *Stannous chloride solution* : 10g of crystalline stannous chloride (LR) is dissolved in 25 ml of concentrated HCl by warming and stored in an amber coloured bottle, carefully avoiding all contact with air. This is 40% $SnCl_2$ stock solution. Just before use, 0.5 ml is diluted to 66 ml with distilled water. A piece of tin metal (AR) added will keep the stock solution for long.

Note : For phosphorus estimation Pyrex glass apparatus (particularly new ones) are to be avoided to minimise contamination with arsenic, which interferes in the analysis. All the glasswares should be cleaned with chromic acid and thoroughly washed.

Procedure

5 ml of soil extract is pipetted into a 25 ml volumetric flask to which 5 ml of the Dickman and Bray's (i or ii) reagent is poured in. In case of Olsen's method, the reagent (ii) should be added drop by drop with constant shaking till the effervescence due to CO_2 evolution ceases. The neck of the flask is washed down and the contents diluted to about 22 ml. Then one ml of the diluted stannous chloride solution is added and volume made up to mark. The intensity of the blue colour is measured (using 660 mμ filter) just after 10 minutes and the concentration of P determined from the standard curve. This is very important as the colour starts fading after some time. With each set of samples a blank (without soil) must be taken. With good quality reagents colorimeter reading for blank is negligible.

Colorimetric estimation of phosphorus by ascorbic acid method (Watanabe and Olsen 1965)

Reagents

(i) 12g of ammonium molybdate (AR) is dissolved in 250 ml of distilled water. In 100 ml of distilled water a solution of 0.29lg of antimony potassium tartrate is prepared separately. Both these solutions are added to 1000 ml of approx. 5N H_2SO_4 (140 ml conc. H_2SO_4 in one litre), mixed thoroughly and made to two litres with distilled water.

(ii) *Ascorbic acid solution* : 1.056g of ascorbic acid (LR) is dissolved in 200 ml of reagent (i) and mixed well. This should be prepared fresh as and when required.

5 ml aliquot of the Olsen's extract is pipetted into a 25 ml volumetric flask and carefully acidified with 5N H_2SO_4 to pH 5. This can be easily done by taking 5 ml of the extracting reagent in a separate 25 ml flask and determining the volume of acid required to bring the pH of the solution to 5 using p-nitrophenol indicator the yellow colour of which disappears at this pH. On adjustment of the pH it is diluted to 20 ml with distilled water and 4 ml of reagent (ii) added. The volume is made up to mark and the contents shaken well. After waiting for 10 minutes the intensity of the blue colour is read in a photoelectric colorimeter using 730-840 mμ filter. A blank must also be run (without soil) alongwith. The colour is relatively more stable than with stannous chloride reduction.

Standard curve for phosphorus

Analytical grade potassium dihydrogen orthophosphate (KH_2PO_4) is dried in air oven at 60°C for 1 hr and after cooling (desiccator) exactly 0.439g is dissolved in about half a litre of distilled water. 25 ml of 7N H_2SO_4 (approx.) is added and made up to one litre with distilled water. This gives a 100 ppm stock solution of P (100 μg P per ml). From this, a 2 ppm P solution is made (50 times dilution). For preparation of the standard curve different concentrations of P (1, 2, 3, 4, 5 and 10 ml of 2 ppm P soln.) are taken in 25 ml volumetric flasks. To these, 5 ml of the extracting reagent (Bray's/Olsen's) is added and the colour is developed as described above by adding Dickman and Bray's reagent (i/ii) and stannous chloride. In case of ascorbic acid method the colour is developed after pH adjustment of the aliquot. The colorimeter reading is taken against 660 mμ (red) filter just after 10 minutes. The curve is plotted taking the colorimeter reading on the vertical axis and the amount of P (in μg) in the horizontal one.

Calculation

Available P (kg/ha)

$$= R \times \frac{\text{Total volume of the extractant}}{\text{Volume of aliquot}} \times \frac{1}{\text{Weight of soil taken}} \times \frac{2.24 \times 10^6}{10^6}$$

where R = μg P in the aliquot (to be seen from the standard curve).

Olsen's P (kg/ha)

$$= R \times \frac{50}{5} \times \frac{1}{2.5} \times 2.24 = \mu g \text{ of } P \times 8.96$$

Bray's P (kg/ha)

$$= R \times \frac{50}{5} \times \frac{1}{5} \times 2.24 = \mu g \text{ of } P \times 4.48$$

VIII. AVAILABLE POTASSIUM DETERMINATION

The term available potassium incorporates both exchangeable and water soluble forms of the nutrient present in soil. The readily exchageable plus water soluble K is determined in the neutral normal ammonium acetate extract of soil. The ammonium ion provides a sharp and rapid separation from exchange complex while other cations bring about a gradual replacement of either lesser or greater amount of K which generally increases with the period of contact. The estimation of potassium in the extract is carried out with the help of flame photometer. Chemical methods being rather elaborate and time-consuming are not suitable for soil testing purpose.

Non-exchangeable K (not lattice-bound) is attributed to contribute appreciably towards the potassium available to crops. The most commonly employed extractants are the strong inorganic acids like 6N H_2SO_4, hot 1N HCl and boiling 1N HNO_3. The procedure for determination involves either the prior removal of exchangeable K or conditions are made sufficiently vigorous to extract both exchangeable (including water soluble) and a portion of non-exchangeable forms from which the former is subtracted.

Instruments

Flame photometer, direct reading type and glass electrode pH meter

Reagents

(i) *Neutral normal ammonium acetate* : Solutions of 2N acetic acid (glacial) and 2N ammonium hydroxide are prepared (by titration with standard alkali and acid) and equal volumes of the two are mixed in a large beaker and on cooling the pH adjusted to 7.0 with acetic acid or ammonia.

(ii) *Potassium chloride solution* : A stock solution of 1000 μg K/ml is made by dissolving 1.908 g of AR grade potassium

chloride (dried at 60°C for 1 hr) in distilled water and making up the volume to 1 litre.

Procedure

5g of soil is shaken with 25 ml of neutral normal ammonium acetate (pH 7) for 5 minutes and filtered immediately through a dry filter paper (Whatman No.1). First few ml of the filtrate may be rejected. Potassium concentration in the extract is determined in the flame photometer after necessary setting and calibration of the instrument.

Standard curve for potassium

From the stock solution, measured aliquots are diluted (in 50 or 100 ml vol. flasks) with the ammonium acetate solution to give 10 to 40 ppm of K. After attaching the appropriate filter and adjusting the gas and air pressure (as per directions given in the operation manual) the flame photometer reading is set at zero for the blank (ammon. acetate) and at 100 for 40 ppm K. The curve is obtained by plotting the readings against the different concentrations (10, 15, 20, 25, 30, 40 ppm) of K. Any fluctuation in gas and air pressure does not allow steady reading in the meter and must be taken care of.

Calculation

Available K (kg/ha)

$$= R \times \frac{\text{Volume of the extract}}{\text{Weight of soil taken}} \times \frac{2.24 \times 10^6}{10^6}$$

where R = ppm of K in the extract (obtained from standard curve)
= ppm of K × 11.2

Procedure for the determination of non-exchangeable potassium
(Wood and DeTurk 1940)

To 5g of the sieved soil (2 mm) placed in a dry 100 ml conical flask 50 ml of 1N HNO_3 is pipetted in and the contents heated on a hot plate (low heat) under reflux (fitted with cork and long glass tube) for 10 minutes after boiling starts. On cooling, the suspension is filtered through dry Whatman No. 1 filter paper and potassium estimated flame photometrically after appropriate dilution. The non-exchangeable K is obtained by subtracting the ammonium acetate K from this value.

IX. LIME REQUIREMENT OF ACID SOIL

While a slightly acidic condition is often favourable for crop growth a higher degree of soil acidity mostly exerts an adverse effect and hence

such acid soils need lime application to improve the productivity. The quantity of lime required to bring the pH to the desired level will vary according to the nature of the soil. Number of laboratory methods are available for this purpose The procedure given by Shoemaker *et al.* (1961) is being widely followed for determining the lime requirement of acid soils of pH less than 6.0. A glass electrode pH meter is the instrument needed.

Reagent

Extractant buffer : 1.8g nitrophenol, 2.5 ml triethanolamine, 3.0g potassium chromate, 2.0g calcium acetate and 53.1g calcium chloride dihydrate (all chemically pure) are dissolved in a litre of water and the pH adjusted to 7.5 with dilute NaOH solution.

Procedure

To 5g of air-dry soil taken in a dry 50 ml beaker, 5 ml distilled water and 10 ml of the extractant buffer are added and stirred continuously for 10 minutes or intermittently for 20 minutes. The pH of the suspension is determined on the basis of which the requirement of lime is read from the following table. The values are given in tons of pure calcium carbonate per acre required to bring the soil to the pH indicated and are to be converted to their equivalents of the form of agricultural lime to be used. For expressing in metric units *i.e.*, tonnes/ha, the figures are to be multiplied by 2.51.

Lime requirement to bring the soil to the desired pH level according to pH value of the soil-buffer suspension

pH of soil-buffer suspension	Lime required to bring the soil to indicated pH (in tons/acre of pure calcium carbonate)		
	pH 6.0	pH 6.4	pH 6.8
6.7	1.0	1.2	1.4
6.6	1.4	1.7	1.9
6.5	1.8	2.2	2.5
6.4	2.3	2.7	3.1
6.3	2.7	3.2	3.7
6.2	3.1	3.7	4.2
6.1	3.5	4.2	4.8
6.0	3.9	4.7	5.4

Table (contd.)

5.9	4.4	5.2	6.0
5.8	4.8	5.7	6.5
5.7	5.2	6.2	7.1
5.6	5.6	6.7	7.7
5.5	6.0	7.2	8.3
5.4	6.5	7.7	8.9
5.3	6.9	8.2	9.4
5.2	7.4	8.6	10.0
5.1	7.8	9.1	10.6
5.0	8.2	9.6	11.2
4.9	8.6	10.1	11.8
4.8	9.1	10.6	12.4

X. GYPSUM REQUIREMENT OF ALKALI SOIL

The reclamation of alkali soil (pH in water suspension exceeding 8.5) requires gypsum treatment for replacement of sodium ions from the exchange complex. The sodium so released has to be leached out by flooding. The gypsum requirement can be determined by adding a known excess of saturated solution of gypsum ($CaSO_4.2H_2O$) to soil and estimating its unreacted amount by EDTA (versenate) titration (Schoonover 1952).

Reagents

(*i*) *Saturated calcium sulphate solution* : About 5g of pure $CaSO_4.2H_2O$ taken in one litre of distilled water shaken mechanically for 10 minutes and filtered

(*ii*) *Ammonium chloride-ammonium hydroxide buffer of pH 10* : 67.5g of pure ammonium chloride dissolved in 570 ml of concentrated ammonia solution (sp. gr. 0.88) and diluted to one litre with distilled water and adjusted to pH 10 (*Caution*: Liq. ammonia should be refrigerated before opening the bottle)

(*iii*) *Eriochrome black T indicator* : 0.5g of eriochrome black T (solochrome black) and 4.5g of hydroxylamine hydrochloride (AR) dissolved in 100 ml of 95% ethyl alcohol

(*iv*) *0.01N calcium chloride solution* : 0.500g of AR grade calcium carbonate taken in little excess of AR HCl (about 10 ml of dil. acid) and the solution made up to one litre with distilled water

(*v*) *Standard versenate (EDTA) solution 0.01N* : 2.0g of disodium dihydrogen ethylene-diamine-tetra acetic acid and 0.05g of $MgCl_2$ (AR) dissolved in water and diluted to 1 litre; the solution to be standardized against 0.01N calcium chloride

Procedure

5g of soil is placed in a 250 ml conical flask to which 100 ml of the saturated $CaSO_4$ solution is poured in, shaken for 5 minutes and filtered (Whatman No.1). After rejecting first few ml, 5 ml of the extract is pipetted into a 100 ml flask or porcelain dish and diluted to about 25 ml with distilled water. One ml of NH_4Cl-NH_4OH buffer and 3 to 4 drops of eriochrome black T indicator are added and titrated with the standard EDTA solution until the colour changes from wine red to blue. Similarly, 5 ml of the saturated $CaSO_4$ solution is titrated separately (blank).

Calculation

Milliequivalent (m.e.) per litre of Ca or Ca+Mg

$$= \frac{\text{ml of the versenate solution used} \times \text{normality of the versenate soln.} \times 1000}{\text{ml of aliquot taken}}$$

Gypsum requirement in m.e. per 100g

= (Ca conc. of added gypsum solution in m.e./l minus Ca+Mg conc. of filterate in m.e./l) $\times 2$

Gypsum requirement : in metric tons (tonnes) per hectare (30 cm soil depth)

= 1.72 × m.e. of gypsum per 100g soil × 2.47

XI. PHOSPHORUS FIXING CHARACTERISTICS OF SOIL

Soils vary appreciably in their phosphorus fixing capacity, those dominating in sesquioxides and kaolinitic type of clays having tendency to fix or immobilize larger amounts of phosphorus from the soluble phase. The phosphorus fixing capacity can be experimentally determined as per procedure outlined by Waugh and Fitts (1966). A double cell photo-electric colorimeter is needed for phosphorus estimation.

Reagent

Standard phosphorus solution : A stock solution of 2500 ppm of P is prepared by dissolving 9.440g of pure anhydrous monocalcium phosphate, $Ca(H_2PO_4)_2$ in distilled water and made to one litre. (Old stock

of MCP may not completely go into solution and should not be used). It is desirable to check the concentration of P by Dickman and Bray's method (1940) after appropriate dilution. Out of this, solutions of different concentrations are prepared (by dilution) such that by adding one ml of each to 2.0g of soil the resultant concentration of added P would be 0, 25, 50, 75, 100, 125, 250, 375 and 500 ppm.

Procedure

2g of soil is weighed out (in duplicate) in separate 100 ml conical flasks. One ml of the solutions of different concentrations of P are added to each so that they cover the range from 0 to 500 ppm as stated above. The flasks are plugged with cotton wool and allowed to incubate for 96 hours at room temperature and thereafter the phosphorus is extracted by Olsen's or Bray's method and estimated colorimetrically (Item VII) as the case may be.

The results of extracted P are plotted (Y-axis) on the graph sheet against added P (X-axis). The curve is examined carefully and the point where there is an abrupt increase in extractable P is located. From this inflexion point (on the curve) the corresponding value of X is found out *i.e.*, approximate amount of P required to be added to more than overcome the effect of fixation. If the fixation curve is linear its shape may be considered to work out the amount of P to be applied which would take into account the fixing capacity.

Calculation (for percentage P fixation)

Amount of available P in control (without P added)	a ppm
Amount of available P in test sample	b ppm
Amount of available P extracted	(b-a)=x ppm
Amount of P added	c ppm
Amount of P fixed	(c-x) ppm
Percentage of P fixation (% of added P)	$\frac{(c-x)}{c} \times 100$

XII. POTASSIUM FIXING CAPACITY OF SOIL

On addition of fertilizer potash to soil, potassium ions from solution readily go into non-exchangeable or fixed form which apparently has a restricted availability to crops. The fixation characteristics differ

according to the soil type, its texture, nature of dominating clay minerals and also in relation to submerged or arable conditions. The procedure given below has been adopted from the All India Coordinated Project on Soil Test and Crop Response Correlation, for finding out the potassium fixing capacity (STCR 1968). A flame photometer (direct reading type) is needed for estimation of potassium in test solutions.

Reagents

(i) Potassium chloride (AR grade) for preparing solution of different concentrations of K

(ii) *Neutral normal ammonium acetate solution*
(Preparation described under item VIII)

Procedure

1. *For sandy to sandy loam soils*

The following enrichment in respect of K is to be adopted :

0, 10, 20, 30, 40, 50, 100, 200, 300, 600 and 1000 ppm K. Soil to solution ratio is 1:2 *i.e.*, 5g/10 ml.

The determination of potassium fixing capacity is carried out for rice and wheat growing soils as is given under (a) and (b), respectively.

(a) Soils growing rice

Standard solution of 1000 ppm K (1.908g of KCl/l) is prepared and diluted so as to give the following concentrations :

Solution A : 10 ppm K (5 ml stock solution diluted to 500 ml)

S lution B : 100 ppm K (50 ml stock solution to 500 ml)

5g samples of sieved soil (2 mm) are taken in duplicate in 100 ml conical flasks. KCl solution and distilled water are added by pipette as given in the table below to give the various levels of exchangeable K.

Levels of exch. K in µg per 5g of soil	KCl solution to be added	Distilled water (ml)
0 (control)	0	10
10	1 ml solution A	9
20	2 ml solution A	8
30	3 ml solution A	7
40	4 ml solution A	6
50	5 ml solution A	5
100	1 ml solution B	9
200	2 ml solution B	8
400	4 ml solution B	6
600	6 ml solution B	4
1000	1 ml stock solution	9

The flasks are stoppered after mixing the contents and equilibriated for 72 hours at room temperature. 25 ml of neutral normal ammonium acetate is added and shaken for five minutes. The suspension is filtered and the K content in the filtrate is determined with the help of flame photometer (Item VIII).

(b) For soils growing wheat

5g of soil is weighed (in duplicate) in porcelain basin and half of the required amounts of KCl and distilled water are added, mixed thoroughly and allowed to dry in shade for 24 hours. After drying, 1/4th amounts of the required KCl and distilled water are mixed well and again allowed to dry for 24 hours. Thereafter the remaining 1/4th quantity of both is added and dried as before. The determination of water soluble and exchangeable K is carried out as given above under rice growing soils.

2. For loam to heavy clay soils

The following enrichment for K is to be adopted :

0, 10, 20, 30, 40, 100, 200, 400, 800, 1000, 2000, 3000 and 4000 ppm K by using *Solution A* : 10 ppm K ; *Solution B* : 100 ppm K ; and *Solution C* : 500 ppm K.

added are :

Levels of exch. K in µg per 5g soil	KCl solution to be added	Distilled water (ml)
0	0	10
10	1 ml of soln. A	9
20	2 ml of soln. A	8
30	3 ml of soln. A	7
40	4 ml of soln. A	6
100	1 ml of soln. B	9
200	2 ml of soln. B	8
400	4 ml of soln. B	6
800	8 ml of soln. B	2
1000	2 ml of soln. C	8
2000	4 ml of soln. C	6
3000	6 ml of soln. C	4
4000	8 ml of soln. C	2

Calculation

The neutral normal ammoniun acetate extractable K in the control sample is deducted from the K content of each treated soil sample. The difference so obtained (Y) is subtracted from the amount of K added (X).

The fixed K (X-Y) is expressed as the percentage of K added.

$$\text{Per cent K fixed} = \frac{(X-Y)}{X} \times 100$$

XIII. SOIL TEST FOR AVAILABLE SULPHUR

Chemical tests for estimation of available sulphur are of recent interest in soil testing work particularly in some reported sulphur deficient areas and for certain crops whose requirement of this nutrient is high enough, often exceeding that of phosphorus.

Of several extracting procedures suggested for available sulphur determination, starting from distilled water for dissolving out the sulphate form to different chemical extractants, the use of phosphate solution has often been preferred for screening the soil for its sulphur fertility status. A number of other reagents have also been tried such as, Olsen's,

Morgan's and Bray's, normal magnesium acetate and 0.15 per cent calcium chloride solution.

As regards the estimation of sulphur in the extract turbidimetric procedure is being most widely followed. The extent of turbidity due to fine suspension of barium sulphate formed by reacting sulphate ions with barium chloride in solution can be accurately measured in the nephelometer (working on the principle of scattered light) but within the range of low concentration spectrophotometer or even colorimeter can work fairly satisfactorily.

Recently, a new colorimetric method of sulphur estimation has been developed in the Soil Testing Laboratory of the I.A.R.I., New Delhi, based on the earlier suggestion of Nemeth (1964). In this procedure, barium chromate reacting with sulphate in solution precipitates the latter as $BaSO_4$ releasing equivalent quantity of chromate ion which is subjected to colorimetric measurement. The method can work quite well for a variety of extractants except those containing phosphate because when the reaction of the medium is changed from acidic to alkaline for removing unreacted barium chromate, barium phosphate also gets precipitated thus affecting the final colour intensity.

Two procedures of extraction and estimation of sulphur (as sulphate) are being described here.

1. *Phosphate extractable sulphur* (Ensminger 1954)

In a 250 ml conical flask, 20g of air-dried processed soil is shaken with 100 ml of 500 ppm P solution prepared from potassium dihydrogen orthophosphate or monocalcium phosphate (KH_2PO_4 2.195 g/litre or $Ca(H_2PO_4)_2$. H_2O 1.888 g/litre) for 30 minutes. The extract is filtered through Whatman No. 42 filter paper and the sulphate determined in a suitable aliquot by turbidimetric procedure.

Turbidimetric method for sulphate estimation

(Massoumi and Cornfield 1963)

Reagents

(*i*) Nitric acid 25% (approx.) prepared from AR grade HNO_3

(*ii*) *Acetic-phosphoric acid* : 900 ml of AR glacial acetic acid mixed with 300 ml of AR orthophosphoric acid

(*iii*) *Gum acacia-acetic acid solution* : 2g of chemically pure gum acacia is dissolved in 500 ml of hot water and the hot solution filtered through Whatman No. 42 filter paper. The filtrate is cooled and then diluted to one litre with dilute acetic acid.

(*iv*) *Barium sulphate seed suspension* : 18g of AR grade barium chloride is dissolved in 44 ml of hot water and 0.5 ml of the concentrated standard sulphate solution (given below), is added brought to boiling and cooled quickly. Then 4 ml of the gum acacia-acetic acid solution is put in. The seed suspension is prepared a fresh every day before use.

(*v*) Barium chloride : $BaCl_2$ analytical grade reagent ground to pass through 1mm sieve.

(*vi*) *Concentrated standard sulphate solution* : (2mg S/ml) : 1.080g of oven-dried AR potassium sulphate dissolved in distilled water and diluted to 100 ml.

(*vii*) *Working standard sulphate solution* (10 µg S/ml) : 5 ml of the concentrated standard solution is diluted to one litre with distilled water.

Procedure

A suitable aliquot of the extract (not more than 15 ml and containing less than 120 µg of sulphate sulphur) is taken in a 25 ml volumetric flask to which 2.5 ml of 25% nitric acid and 2 ml of acetic-phosphoric acid are added and diluted to about 22 ml. The flask is stoppered and shaken. Then 0.5 ml of barium sulphate seed suspension (which must be shaken just before use) and 0.2 g of barium chloride crystals are added successively. The flask is stoppered and inverted three times. After 10 minutes it is inverted 10 times and after 5 minutes another 5 times. Allowing for another 5 minutes, 1 ml of gum acacia-acetic acid solution is put in and diluted to volume and inverted 3 times and set aside for 1½ hr. Then the flask is again inverted 10 times and reading taken in Klett-Summerson colorimeter against blue filter (optimum transmission at approximately 440 mµ) or in spectrophotometer. The concentration of sulphur is found out from the standard curve.

Preparation of standard curve for Sulphate Sulphur

0, 1, 3, 5, 8, 10 and 12 ml of the working standard sulphate solution are placed in separate 25 ml volumetric flasks and proceeded as described above.

The standards are to be prepared each time a batch of sample is analysed and also a blank test with the reagents is to be run side be side.

2. Colorimetric procedure for available Sulphur estimation using Barium Chromate (Palaskar *et al.* 1981)

The sulphur is extracted in 0.15% $CaCl_2$. $2H_2O$ solution in 1 : 5 soil to solution ratio using 10-20 g of soil and 30 minutes shaking time (Williams and Steinbergs 1959). Sulphur is estimated in a suitable aliquot (5-10 ml) of the filtered (Whatman No. 42) extract.

Reagents

(i) *Barium chromate solution* : AR grade 0.1N in 1.5N HCl

(ii) Ammonium hydroxide : 5N (approx.)

(iii) *Standard sulphate solution* : (100 ppm S) prepared from AR grade potassium sulphate

Procedure

5 ml of barium chromate solution is pipetted into 100 ml volumetric flask to which 1.2 ml of 5N ammonium hydroxide is added in order to reduce the free acidity to about 0. 05N. No turbidity should appear. Measured aliquot of soil extract is poured in so as to keep the sulphate sulphur content in the flask below 2000 μg. After shaking and standing for about half an hour 1 ml of ammonium hydroxide is added to precipitate the unreacted barium chromate. The volume is made up to 100 ml with distilled water. The flask is stoppered and inverted 2-3 times and the content is filtered through dry Whatman No. 42 filter paper. rejecting the first few ml of filtrate. The intensity of yellow colour in the filtrate is measured in photoelectric colorimeter using deep blue (420 mμ) filter, and the concentration found out from the standard curve.

It is necessary that the test solution should have a *pH* around 2 to 3 before barium chromate is added. In case of alkaline extract the *pH* has to be brought down with required amount of 6N HCl.

Preparation of standard curve

From 100 ppm working solution of sulphate sulphur, 0, 4, 8, 16, and 20 ml are taken into 100 ml volumetric flasks and the colour intensity read as given above. *Blank correction* is to be made.

precipitated out with ferric salt ($FeCl_3$) and ammonia during the same operation of removal of precipitated barium sulphate and chromate but if excessive amount of phosphate is present in the test solution turbidimetric estimation is recommended.

XIV. ESTIMATION OF AVAILABLE ZINC

Soil test for micronutrient elements is still at the developing stage. Of late, their quick estimation has become possible with the help of atomic absorption spectrophotometer. In recent years, zinc has been receiving much attention as there are several reports of its deficiency and significant crop response to zinc application. To some extent, iron deficiency has also been observed in certain areas while deficiency of manganese and copper has been reported to a much lesser extent.

Several reagents have been employed from time to time for the extraction of available micronutrients in soil out of which DTPA appears as quite promising having a wider applicability. Four micronutrient elements *viz*, zinc, iron, manganese and copper can be estimated in a single extraction with DTPA by setting the instrument for each of them and using the respective standard curve.

Note : Adequate precautions must be observed against any likely contamination from the reagents and glasswares in micronutrient assay work. Only double distilled water should be used.

Determination of DTPA extractable Zinc

(Lindsay and Norvell 1969, 1978)

Instrument

Atomic absorption spectrophotometer (AAS)

Reagents

(i) *DTPA* (Diethylene-triamine penta acetic acid) 0.005M solution

(ii) *TEA* (Triethanolamine) 0.1M (AR or extra pure)

(iii) $CaCl_2$. $2H_2O$ (AR) 0.01M solution

(iv) Zinc metal (AR or guaranteed reagent)

(v) Dilute HCl (AR HCl diluted with double distilled water 1 : 5)

1.470g $CaCl_2.2H_2O$ in a beaker. To this 20 to 25 ml of double distilled water is added and 13.3 ml of TEA followed by 100 ml of double distilled water. This is transferred to one litre volumetric flask with 3-4 washings and the volume made up to mark with water. The *pH* of the solution is adjusted to 7.3 with dilute HCl.

Procedure

To 10g of soil in 100 ml conical flask, 20 ml DTPA reagent is added and shaken for two hours. The extract is filtered (Whatman No. 40/42) and zinc estimated with the help of atomic absorption spectrophotometer.

Standard curve for Zinc

One gram of pure zinc metal is dissolved completely in minimum amount of dilute HCl and made up to one litre with double distilled water in an already used (not new) volumetric flask (Corning/Pyrex) and then transferred to a plastic bottle. This stock solution contains 1000 µg Zn/ml. A working standard of 25 µg/ml is prepared by diluting 2.5 ml of the stock solution to 100 ml.

In six 100 ml volumetric flasks required aliquots are taken and standards of 0, 0.5, 1.0, 1.5, 2.0 and 2.5 ppm zinc solution are made and the standard curve prepared against the readings of the atomic absorption spectrophotometer after necessary setting and calibration of the instrument (AAS).

Calculation

In the same manner as given in case of potassium (item VIII) after taking into account the dilution factors.

XV. DETERMINATION OF AVAILABLE COPPER, IRON AND MANGANESE

A mention has been made about the appearance of copper, iron and manganese deficiency in certain places. At the same time, toxicity due to excessive occurrence of some elements particularly iron and manganese has also been brought to light in acidic soils. Not much attempt, however, has been made to estimate the available boron, molybdenum

toxicity. The following method has found favour for the estimation of copper, iron and manganese over the colorimetric procedures (Lindsay & Norvell 1978).

The same DTPA extract of soil obtained for the estimation of available Zn can be utilized for determination of available Cu, Fe and Mn as well. A separate standard curve for each element is prepared and the appropriate (specific) hollow cathode lamp used on the AAS. The required standard solutions are prepared as follows;

Copper

A stock solution of 1000 ppm Cu is obtained by dissolving exactly one gram of the pure (AR grade) metal (wire or turning) in 50 ml of dilute (1 : 1) HNO_3 and finally diluting to 1000 ml with double (glass) distilled water. From this, working solutions (0.25, 0.5, 1.5, 2.0, 2.5 and 3 ppm) are prepared (by dilution) in 100 ml volumetric flasks.

Iron

To prepare a stock solution of 1000 ppm Fe, exactly 1g of AR grade Fe metal is dissolved in 50 ml of dilute HNO_3 and made to one litre with double distilled water. Working solutions (1, 2, 4, 6, 8, and 10 ppm) are made in 100 ml volumetric flasks by appropriate dilution.

Manganese

1.583g of MnO_2 (AR) or 1.0g of pure Mn metal dissolved in 50 ml of dilute HNO_3 and made up to one litre gives a stock solution of 1000 ppm Mn. In 100 ml volumetric flasks requisite quantities of the stock solution are diluted to obtain working concentrations of 1, 2, 4, 6, 8 and 10 ppm Mn.

The amounts of these nutrients in soil are calculated in a similar manner as for available potassium (Item VIII).

Section B

QUALITY RATING OF IRRIGATION WATER

Water is the most universal solvent and its all naturally occurring sources are bound to contain variable quantities of dissolved solids and gases. Sometimes, suspended and colloidal organic and inorganic materials occur as well. Different types of laboratory tests of water are carried out depending on its use from the point of view of industry, public health or agriculture. Thus a water considered to be of poor quality for industrial or domestic purpose may be quite suitable for irrigation in which case total soluble salts, nature and concentration of various ions and particularly the proportion of the divalent and monovalent cations are of specific relevance in judging the water quality.

Collection of irrigation Water Sample

The sample of irrigation water is collected in about 500 ml glass or polythene bottles (stoppered) which should preferably be transparent. The container must be thoroughly cleaned before use and should be rinsed 3 to 4 times with the water from which the sample is to be drawn. Brand new bottles are better to avoid and when used the cleaning and washing have to be more rigorous after keeping these filled with water for one or two days.

If the source of irrigation water is a tank, canal or river the sample should be drawn either from a spot away from the sides or from the mid-stream. This can be easily managed with the help of a small bucket tied at the end of a long pole. The tubewell or hand pump is to be run for about 10 minutes prior to sampling and for open well, several buckets of water have to be thrown out first. About half a litre of the sample is quite sufficient. The water sample after proper marking and labelling must be sent to the laboratory *immediately* for testing to avoid any change or deterioration. If a few days delay is inevitable then 2 or 3 drops of pure toluene may be added to prevent bacterial activity.

The water sample is straightway taken for analysis. Any little amount of sediment if present may be allowed to settle down. In very few cases it becomes necessary to filter the water (Whatman No. 1) for testing purpose.

1. Determination of *pH*

The *pH* is determined by taking about 50 ml of the water sample in a 100 ml clean beaker and immersing the electrodes of the pH meter as described under soil analysis (Item II, Section A).

2. Electrical Conductivity measurement (for Soluble Salts)

The cell of the conductivity meter or solubridge is filled with the water sample and the electrical conductivity (EC) is measured as in case of water extract or clear suspension of soil (Item III) and expressed as mmhos/cm at 25°C (sometimes the unit of expression is micrommhos/cm that is 1/1000th of millimhos/cm when the EC is rather low),

3. Estimation of Calcium and Magnesium

The usual method for the determination of Ca+Mg is by versenate (EDTA) titration (Cheng and Bray 1951).

Reagents

(i) *Standard versenate solution* : An approximately 0.01N solution of ethylene diamine tetra-acetic acid disodium salt (versenate) is prepared by dissolving 2.0g in distilled water to which 0.05g of magnesium chloride ($MgCl_2.6H_20$) is added and diluted to one litre. This is to be standardized against 0.01N calcium chloride solution prepared by accurately weighing out 0.500g AR grade $CaCO_3$ (oven dried) and dissolving in minimum excess of dilute HCl (AR) followed by making up to one litre with distilled water.

(ii) *Ammonium chloride-ammonium hydroxide buffer (pH 10)* : 67.5g pure ammonium chloride dissolved in 570 ml of concentrated ammonium hydroxide and made to one litre and adjusted to pH 10. (*Caution* : Ammonia bottle must be titration (or to the temperature before opening). ith phenolphthalein) and the titration con-

(iii) *Eriochrome black T* (*Solochrome black*) *indicator* : 0.5g of eriochrome black T and 4.5g of hydroxylamine hydrochloride (AR) dissolved in 100 ml of 95% ethyl alcohol.

Procedure

5 ml of the water sample in a porcelain dish (8 cm dia.) is diluted to about 25ml with distilled water. One ml of ammonium chloride-hydroxide buffer and 3 to 4 drops of eriochrome black T indicator are added and titrated with the standard versenate solution. The colour change is from wine red to blue or bluish green. At the end point no tinge of the red colour should remain.

Calculation

From the volume of 0.01N EDTA (standardized against 0.01N $CaCl_2$) solution required for titration the concentration of Ca+Mg is directly obtained in m.e./litre as follows :

$$\text{m.e./l of Ca+Mg} = \text{ml of versenate soln used} \times \text{normality of versenate solution} \times \frac{100}{\text{ml of aliquot taken}}$$

4. Determination of Sodium and Potassium

Small amounts of sodium and potassium are generally present even in the best quality irrigation water. The concentration of sodium may be quite high in saline water with EC greater than 1 mmhos/cm and containing relatively less amount of Ca and Mg. The concentration of potassium is generally low. Obviously, these estimations are of interest when the water sample tests saline (*i.e.*, having EC above 1.0 mmhos/cm at 25°C).

Instrument

The determinations of Na and K are carried out directly with the help of flame photometer using appropriate filters and standard curves prepared by taking known concentrations of Na and K.

Standard curve for Sodium

As the working concentration of sodium in flame photometric must be sent to elatively higher than that of potassium the strength of change or deterior conveniently expressed in milliequivalents per litre drops of pure toluen 845g of NaCl (AR) dissolved in water and made to

1.058 litre gives 100 m.e./l of Na. From this, solutions having 5, 10, 20, 30, 40 and 50 m.e. Na/l are prepared. A curve is drawn by plotting flame photometer readings on the Y-axis against concentrations of Na on X-axis. The concentration of Na in the unknown sample is read from the curve.

Standard curve for Potassium

A stock solution of 1000 ppm K prepared in distilled water contains 1.908g of AR grade KCl per litre. Dilute solutions having 2, 5, 10, 15 and 25 ppm K are obtained from the above in 100 ml volumetric flasks. Standard curve is prepared by plotting flame photometer readings against concentrations of K.

$$\text{K (m.e./litre)} = \text{ppm K (from the curve)} \div 39$$

$$\text{or, ppm K} \times 0.0256$$

5. Carbonates and Bicarbonates

(Richards 1954)

The estimation is based on simple acidimetric titration using different indicators which work in alkaline pH range (above 8.2) or in acidic *pH* (below 6.0).

Reagents

(i) *Phenolphthalein indicator* : 0.25% solution in 60% ethyl alcohol

(ii) *Methyl orange indicator* : 0.5% solution in 95% alcohol

(iii) *Standard sulphuric acid* (0.01N)

Procedure

In a porcelain dish 5 ml of the water sample (containing not more than one milliequivalent of carbonate plus bicarbonate) is diluted with distilled water to about 25 ml. A pink colour produced with a few (2 to 3) drops of phenolphthalein indicates presence of carbonate and it is titrated with 0.01N sulphuric acid until the colour just disappears (phenolphthalein end point) because of alkali carbonate being converted to bicarbonate. This burette reading (volume used) is designated as V. To the colourless solution from this titration (or to the original sample of water if there was no colour with phenolphthalein) 1 to 2 drops of methyl orange indicator are added and the titration con-

tinued with brisk stirring to the methyl orange end point (yellow) and the final reading noted which is marked as Z.

Calculation

(a) Carbonates (milliequivalents/litre)

$$=2Y \times \text{normality of } H_2SO_4 \times \frac{1000}{\text{ml of aliquot}} = 2Y \times 2 \text{ or } 4Y$$

(b) Bicarbonates (m.e./litre)

$$=(Z-2Y) \times \text{normality of } H_2SO_4 \times \frac{1000}{\text{ml of aliquot}}$$

$$=(Z-2Y) \times 2$$

In case carbonate is absent : $Z \times 2$

6. Residual Sodium Carbonate (RSC)

This is an important character for assessing the suitability of irrigation water in consideration of likely sodium hazard. It is calculated from the analysis data for carbonates, bicarbonates and calcium plus magnesium in the following manner :

$$RSC\ (m.e./l) = (CO_3^{=} + HCO_3^{-}) - (Ca^{2+} + Mg^{2+})$$

all expressed in m.e./l.

The classification of irrigation water according to RSC is given below :

Residual sodium carbonate (RSC) (m.e./l)	Interpretation regarding suitability of the water from the point of view of sodium hazard.
Less than 1.25	Can be safely used for irrigation
1.25 to 2.50	Marginal (to be used with due caution)
Above 2.50	Unsuitable

7. Determination of Chloride

Chloride is invariably present in small amounts in almost all natural waters and its content goes up appreciably with increasing salinity.

This estimation (by Mohr's titration) may be carried out when the electrical conductivity of the water sample is greater than 1 mmhos/cm at 25°C.

Reagents

(i) *0.02N sodium chloride* : 1.170g of NaCl (AR grade, dried at 80°C for 1 hour) is dissolved in double distilled water and made to one litre

(ii) *0.02N silver nitrate* : 3.40g of silver nitrate is dissolved in double distilled water and made up to one litre. This to be standardized against the standard NaCl solution and stored in amber coloured bottle **away from light**

(iii) *Potassium chromate indicator* : 5% aqueous solution of pure K_2CrO_4

Procedure

5 ml of the sample (the one after carbonate/bicarbonate titration can also be used) is taken in a porcelain dish and diluted to about 25 ml with water. After adding 5 to 6 drops of K_2CrO_4 indicator it is titrated with the standard $AgNO_3$ solution (with stirring) till the first brick red tinge appears.

Calculation

1 ml of 0.02N $AgNO_3$ :: 0.00071g of chloride

Chloride in m.e./litre $= V \times 4$

where V is the volume of 0.02N $AgNO_3$ used in titration.

8. Estimation of Nitrate in Irrigation Water

Beneficial effect of nitrates on crop production has been reported specially in brackish waters. The presence of K and NO_3 ions in appreciable amounts has been found to partially counteract the effect of salinity and sodium hazards of irrigation water on crop growth.

River or canal waters unless contaminated with sewage, effluent, city or industrial wastes, generally contain quite low amount of nitrate. Some groundwaters may have appreciable nitrate concentration.

Details of the instrument employed, reagents required and preparation of standard curve have been given earlier under testing of soil extract for nitrate nitrogen (Item VI, Section A).

Procedure

A suitable volume (10-50 ml depending upon the nitrate content) of the water sample is evaporated to dryness over water bath in a 50 ml beaker or dish. On cooling, the yellow colour is developed with phenol disulphonic acid followed by addition of ammonia and the nitrate concentration determined colorimetrically as described under item VI relating to soil testing.

9. Estimation of Sulphate

While traces of sulphate occur universally its content can be appreciable in most saline waters showing EC values more than 1 mmhos/cm (25°C).

Sulphates in solution are quantitatively precipitated in HCl medium and estimated gravimetrically as $BaSO_4$. The sulphate content in many irrigation waters may be quite low and therefore a large volume of the sample has to be first concentrated by evaporating to about 100 ml. The gravimetric procedure itself is time consuming. Turbidimetric and colorimetric methods of determination of sulphate have been described earlier (ItemXIII under soil testing) and the same are also applicable for water analysis. The procedure given here is based on EDTA titration as described by Jackson (1973).

Reagents

(i) *0.02N $MgCl_2$. $BaCl_2$ and EDTA solutions* : While AR grade $BaCl_2$ can be directly weighed out and dissolved, $MgCl_2$ being highly hygroscopic, the required quantity of AR magnesium metal or $MgCO_3$ may be dissolved in little excess of dilute HCl and made up to volume. The EDTA solution must be standardized against 0.02N $CaCl_2$ (Item 3 in water analysis).

(ii) *Buffer solution* : This consists of 8.2 g of NH_4Cl plus 115 ml of NH_4OH (sp. gr. 0.88) in a litre. The amount of NH_4OH may be so adjusted that 10 ml of this solution added to 50 ml of water sample can give a *pH* of 10.

(iii) *Standard calcium solution* : A 0.02N solution is prepared by dissolving 1.001g dried $CaCO_3$ (AR) in minimum quantity of dilute HCl and made up to one litre with water.

(iv) *Eriochrome black T indicator* : 0.5g of the indicator and 4.5g of hydoxylamine hydrochloride (AR) dissolved in 100 ml of 95% ethyl alcohol

Procedure

The $Ca^{2+} + Mg^{2+}$ content of the sample is obtained by direct titration with EDTA (Item 3 section B). In another aliquot, $CO_3^{=} + HCO_3^{-}$ content is found out by titration with standard HCl (0.02N). To a third aliquot, a measured volume of standard HCl equivalent to the total alkalinity ($HCO_3^{-} + CO_3^{=}$) with a slight excess is added and boiled to destroy carbonates and drive off CO_2. Then a known volume of standard $BaCl_2$ is run into this and boiled for a few seconds. On cooling, 10 ml of buffer and 5 drops of eriochrome black T indicator are added and titrated with the standard EDTA solution to the first end point. This is not taken as final as the accuracy is rather poor even if the versenate solution contains some Mg^{2+}. Therefore, after adding a known quantity of standard $MgCl_2$ solution, a second end point is reached with EDTA titration as followed for Ca+Mg determination.

Calculation

Milliequivalents of sulphate = B+Ba+Mg−T in which B is blank (i.e. m.e. of Ca+Mg in the original sample, Ba is m.e. of barium, Mg is m.e. of magnesium added and T is the milliequivalent of versenate (EDTA) required in the total titration of the sample after adding Ba and Mg.

10. Boron in Irrigation Water

The range between the limits of deficiency and toxicity of boron (an essential micronutrient element) is very narrow. Some irrigation waters particularly those from entrapped underground sources or springs may contain excessive amount of boron which is very harmful for plant growth. Thus, a water otherwise testing alright in respect of *pH*, EC, RSC, etc. can be quite unsuitable for irrigation because of likely boron hazard. In all such cases, estimation of boron must be carried out.

Of the few selective organic compounds that form coloured complexes with boron (normally present as borate), the use of curcumin,

boron dependent colour developed with this. The reagent forms coloured lakes with Be, Fe, Al or Mg in alkaline condition and so an acid medium is to be maintained. Interference due to nitrates (when more than 90 ppm) can be eliminated by evaporating the water to dryness after adding 10 ml of saturated solution of pure $Ca(OH)_2$ followed by ignition and extraction with 0.05N HCl (Jackson 1973).

Special precaution regarding the use of glasswares : Since hard or resistant glass is made of boro-silicates, **such glasswares must not be used at any stage,** unless specially made *boron-free*. Ordinary soda glass, silica or polythene wares are quite suitable for boron estimation

Reagents

(i) *Ethyl alcohol 95%* : Distilled in *boron-free glass* or in metallic assembly and stored in soda glass or polyethylene bottles.

(ii) *Curcumin-oxalic acid* : 0.04g of finely ground pure curcumin and 5.0g of AR oxalic acid are dissolved in 95% ethyl alcohol (i) and the solution is stored in amber coloured soda glass bottle in a cool and dark place. A fresh reagent must be prepared after every few days.

(iii) *Standard boron solution* : A stock solution of 100 ppm B is obtained by dissolving 0.572g of dried AR grade boric acid (H_3BO_3) in distilled water (*not glass distilled*) and making the volume to one litre. A working solution of 10 ppm B is prepared from this by dilution.

Procedure

A suitable aliquot of the water sample (containing 0.2 to 5 μg of B) which is made slightly acidic (with 0.05N HCl) is thoroughly mixed with 4 ml of curcumin-oxalic acid reagent in a 250 ml beaker and evaporated to dryness at 55°-60°C on a hot plate or water bath. It is further baked for 15 minutes at this temperature. After cooling, the residue is triturated with 25 ml of 95% ethyl alcohol to extract the rosecynanine coloured complex and filtered (Whatman No. 1) directly into the colorimeter tube. The colour intensity is read spectrophotometrically at 540 mμ and within a *maximum period* of 2 hours.

In separate 250 ml beakers, 0, 1, 2, 3, 4, 6, 8 and 10 ml of the 10 ppm B solution are transferred and the colour developed and measured as stated above. The spectrophotometric/colorimeter readings are plotted on the Y-axis and concentrations of B on the X-axis to obtain the standard curve.

Calculation

Boron (ppm) in water

$$= \frac{\mu g \text{ B read from the standard curve}}{\text{ml of water sample taken}}$$

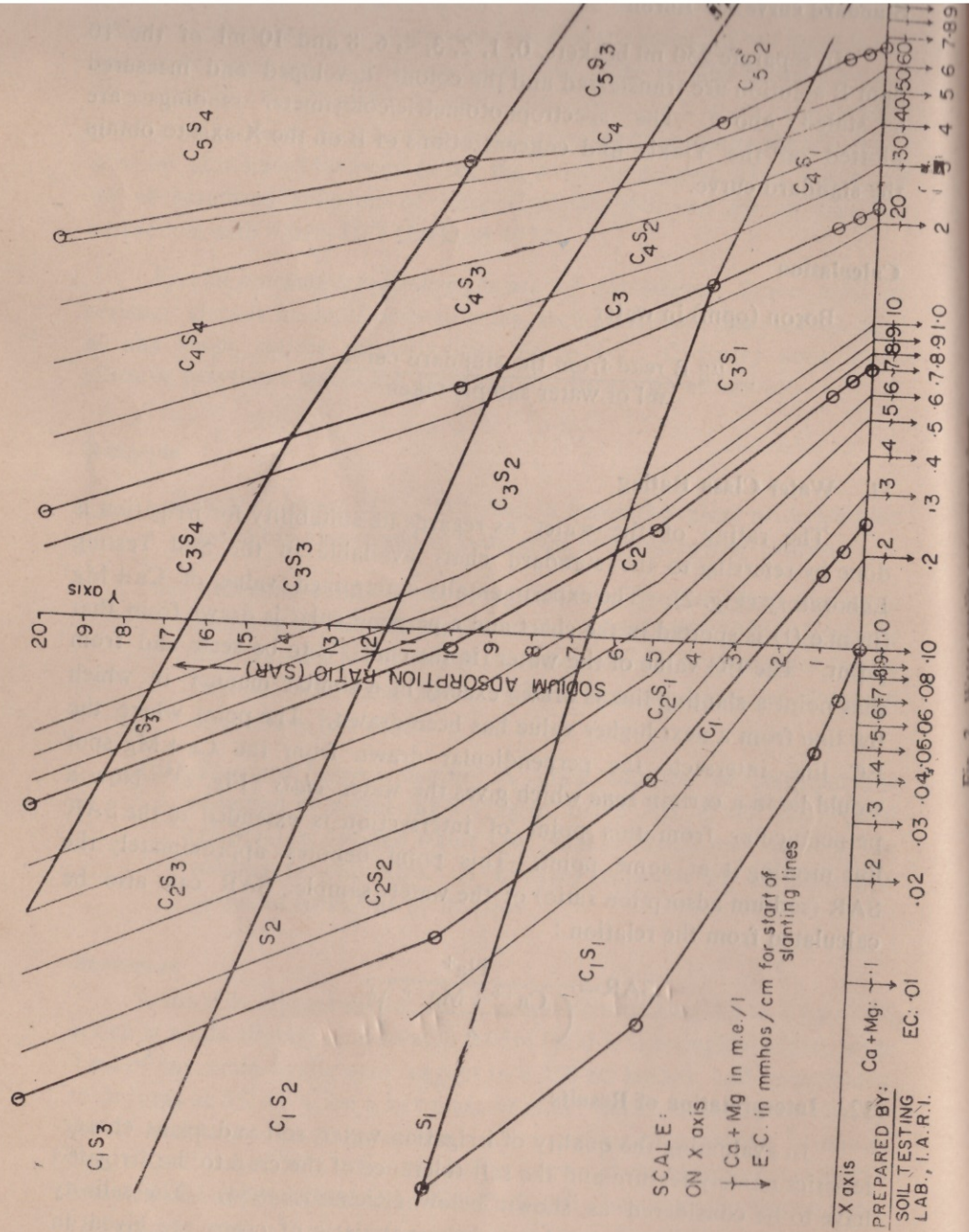
11. Water Class Rating

The rating of the water as regards its suitability for irrigation is done by referring to the standard chart available in the Soil Testing Laboratory (Fig. 2). The experimentally determined value of Ca+Mg (in m.e./l) is spotted in the chart and a perpendicular is drawn from that point. The EC value of the water (in the chart) is to be seen and from that point a slanting line is drawn exactly in a similar manner in which the line from a next higher value has been drawn. The point where the EC line intersects the perpendicular drawn from the Ca+Mg spot would lie in a certain *zone* which gives the water *class* (Fig. 2). Now a perpendicular from this point of intersection is extended to the SAR line meeting it at some point. This point denotes approximately the SAR (sodium adsorption ratio) of the water sample. SAR can also be calculated from the relation :

$$SAR = \frac{Na^+}{\left(\frac{Ca^{++}+Mg^{++}}{2}\right)^{\frac{1}{2}}}$$

12. Interpretation of Results

In evaluating the quality of irrigation water, soil and plant characteristics namely texture and the salt tolerance of the crop to be irrigated have to be considered as shown below (Richards 1954). The salinity appraisal of soils and salt tolerance characteristic of crops are given in appendices V to VII.



Class	Soil texture	Rating	Class	Salt tolerance of crops	Rating
T_1	Clay	4.0	P_1	Sensitive	3
T_2	Clay loam	3.5	P_2	Semi-salt tolerant	2
T_3	Loam	3.0			
T_4	Sandy loam	2.5	P_3	Salt tolerant	1
T_5	Sandy	2.0			

When the irrigation water does not contain appreciable amounts of nitrates, its safe use can be made so long as the sum total of all the indices of soil texture, plant tolerance and sodium and salinity hazards class number (water class) does not *exceed* 9.

The following *example* of testing a water sample (from Delhi state) as regards its suitability for irrigation will help to understand the use of the rating indices :

	Test results
pH	7.5
E.C. (mmhos/cm at 25°C)	2.0
Ca+Mg (m.e./l)	8.6
Carbonates (-do-)	Nil
Bicarbonates (-do-)	8.0
RSC (-do-)	Nil
SAR	5.5
Water class (from fig.2)	C_3S_2

On sandy soils (T_5) the textural rating (given in the table on the top) is 2.0. Using the concept of total of 9, the above water test results work out to give a figure of :

$C_3S_2+(T_5)=3+2+2=7$

rence of 2 (9 – 7=2), the water can be safely used for the two categories of crops in the ascending order *i.e.*, P_3 (salt tolerant) and P_2 (semi-salt tolerant). However, the same water sample on a loam (T_3) soil will be suitable for salt tolerant crops (P_3) only (3+2+3=8 and 9−8=1). Similarly waters having the *sum total* of various ratings equal to or greater than 9, can not be recommended for any type of crop. A dilution in 1:1 ratio with good quality irrigation water may permit use of such water for salt tolerant crops only.

This, of course, is just a *general guideline* and subject to adjustment according to local conditions particularly in relation to soil and plant characterisation.

Section C

INSTRUMENTS IN THE SOIL TESTING LABORATORY

The analytical work in a soil testing laboratory mainly involves standard chemical methods, *suitably modified* to permit handling of a large number of soil samples with the required degree of *accuracy and speed.* Many of the analytical operations are carried out more conveniently with the help of common as well as (moderately) sophisticated instruments.

The various instrumental measurements followed in soil and water testing work in the laboratory can be broadly divided into two groups :

(i) Determination of pH and electrical conductivity ; and

(ii) colorimetry, spectrophotometry, flame photometry and atomic absorption spectrophotometry.

1. pH

The *pH* of a solution, a term introduced by Sørensen has been defined as the negative logarithm of the hydrogen ion *activity* which in very *dilute* solutions can be expressed as concentration, in gram mole per litre.

$$pH = -\log_{10}[H^+]$$

Water even in the purest form dissociates into its component hydrogen and hydroxyl ions, but very *slightly*. The amount of undissociated water is so large that its concentration can be regarded as *constant.* Therefore, in the equilibrium,

$$H_2O \rightleftharpoons H^+ + OH^-$$

the dissociation constant (K) is given by

$$K = \frac{[H^+] \times [OH^-]}{[H_2O]}$$

Since the denominator (concentration of undissociated water) is virtually a constant the product $[H^+] \times [OH^-]$ is also a constant at a given temperature which is equal to 10^{-14} at 25°C or pKw=14.0.

At the neutral point, $[H^+]=[OH^-]=10^{-7}$ g n ole per litre. Thus *pH* at neutrality is $-\log_{10}(10^{-7})$ that is=7. Some points on the Sørensen scale of *pH* are given below :

pH	$[H^+]$	$[OH^-]$
0	$10^{0}(=1)$	10^{-14}
3	10^{-3}	10^{-11}
7	10^{-7}	10^{-7}
10	10^{-10}	10^{-4}
14	10^{-14}	$10^{0}(=1)$

The electrometric determination of *pH* by a pH meter is based on measuring the e.m.f. (electromotive force) of a pH cell with a reference buffer and then with a test solution. The pH cell consists of an electrode sensitive to hydrogen ions (platinum, glass, quinhydrone etc.) and a reference electrode, and can be broadly represented as :

H^+ sensitive electrode	Reference buffer or test solution	Salt bridge	Reference electrode

In the conventional instruments, the measuring electrode is of glass and the reference one is a calomel electrode.

Note : Since the extent or degree of dissociation (here of water) is *temperature dependent*, this pH scale (0-14) is *valid* for a particular temperature only. For other temperatures, necessary adjustments for any difference (in temperature) are to be made.

Glass electrode : This consists of a very thin-walled bulb of H^+ responsive (soft) glass sealed to a stem of non-H^+ responsive glass. Both surfaces of the glass membrane are sensitive to changes in H^+ ion concentration. The inside of the bulb is filled with an electrolyte of high buffer capacity for example, a 0.1N solution of hydrochloric acid. An internal reference electrode, generally a calomel one, also dips into the test or buffer solution.

When the glass bulb is dipped in a solution with [illegible] concentration different from that of the solution within the electrode its outer and inner surfaces attain a difference in electrical potential. This difference is a function of the hydrogen ion concentration of the outer solution, since the *pH* of the inner one is constant (remains unchanged).

Calomel electrode : The electrode consists of mercury in contact with a solution of potassium chloride saturated with mercurous chloride (Hg_2Cl_2). It maintains a *constant potential* at a given temperature. In commercial models, a paste of mercury and Hg_2Cl_2 is contained in an inner tube connected to the KCl solution in an outer jacket. The lead wire is connected to the Hg_2Cl_2 paste through a column of mercury. The outer tube ends in a fine capillary to provide a salt bridge through the test solution to the glass electrode. pH meter with *single* (actually *combined*) electrode is also available as in case of digital type instrument.

Measurement of *pH*

A pH meter is actually a direct current amplifier that measures the e.m.f. which appears across the electrodes upon being immersed in a solution, soil suspension or irrigation water. The glass and calomel electrodes are immersed in the test solution and the e.m.f. determined by an electron tube voltmeter. The meter is graduated to read directly in *pH* units alongwith the e.m.f. (milli-volts) scale. A standard buffer solution (of known *pH*) is used to calibrate the instrument before determining the *pH* of test solutions.

The measurement is registered on a suitably calibrated galvanometer or milliammeter. The circuits of pH meters used in the soil testing laboratories are of high impedence type. Before taking the measurement, it is essential to set the instrument at the temperature of the test solution with the help of the temperature setting knob provided.

Types of pH Meter

A pH meter can be one of the four types :

(i) *Potentiometric or slide type* : The glass electrode is incorporated in an ordinary potentiometric circuit, and the off-balance currents are amplified electronically so that a milliammeter can be used to detect the balance point. The electronic amplifier thus serves only as a null-point indicator.

(ii) *Direct reading* : The e.m.f. of the glass electrode assembly is impressed across a high resistance and the direct current flowing through the resistance is amplified and then passed through a milliammeter ; the deflection indicates the *pH* directly. It is basically a vacuum-tube type voltmeter. The direct reading instrument is characterised by greater rapidity of reading and is readily adopted for A.C. operation.

(iii) *Vacuum tube voltmeter (VTVM)* : VTVM with indicating scales in pH values is calibrated in voltage units for a glass reference electrode pair on the basis of the relationship for the e.m.f. of a pH cell. The apparent e.m.f./pH slope will be 59.15 mV per pH unit at 25°C using the equation pH=pHs+ (E-Es) /0.000198T, where pHs and Es are the values in the standard state, and T is the absolute temperature (in degrees Kelvin).

(iv) *Balanced circuit* : The circuit is balanced initially with the slide wire contactor at *pH* 7 and connected directly to the grid of the electron tube according to the relationship pH=pHs+ (E−Es)/0.000198T, where pHs refer to the pH of a reference buffer solution. Instruments of this type are compact and portable and comparatively less expensive. Often the reading scale extends from 2 to 12 pH units with a reproducibility of 0.1 unit. An example of the more sophisticated form of this type is the Beckman (U.S.A.) model G pH meter.

Most of the pH meters used in the soil testing laboratories in India are vacuum tube voltmeters.

Care and maintenance : The *most delicate part* of the pH meter is the glass electrode. It may crack or break if handled roughly or it may dry up when left out of water for a long period. In the latter case the electrode should be immersed in 0.1N HCl and then in distilled water for 24 hours or more and checked again. To identify defective ones, the electrode jacks are short circuited with the special strap and the terminal connections are provided with the instrument. The pH meter is switched on and sufficient time (10-15 min.) is given for warming up. If the galvanometer needle reads *pH* of 7 or can be adjusted to *pH* 7 with the standardization control in either *pH* range, the electronic side of the pH meter is in order and the defect is in the electrodes and hence these may need replacement.

It is equally essential to see that there is always some solid KCl in the reference (calomel) electrode. The film of clay gels deposited on the electrode needs to be removed with the help of swab of cotton and distilled water.

2. *ELECTRICAL CONDUCTIVITY MEASUREMENT*

Ions being the carrier of electricity, electrical conductivity is a measure of the ability of a salt solution to carry electric current by the migration of ions under the influence of an electric field. Like a metallic conductor, solutions also obey Ohm's law. Increase in temperature *promotes dissociation* with a consequent *rise* in conductivity. The unit of specific conductance is the *reciprocal* of specific resistance in ohms/cm that is, mhos/cm.

If Cs is the concentration of the solution in gram equivalents per litre, then the volume of solution in cubic centimeters per equivalent is equal to 1000/Cs, so that conductance $\Lambda = 1000 \frac{K}{Cs}$ where K is the cell constant. At *infinite dilution*, the ions theoretically are independent of each other and each ion contributes its part to total conductance. Thus

$$\Lambda \propto = \Sigma(\lambda+) + \Sigma(\lambda-)$$

where $\Lambda \propto$ is the total and $\lambda+$ and $\lambda-$ are the ionic conductances of cations and anions, respectively at infinite dilution.

Determination of cell constant : The cell constant K of a conductivity cell is determined by measurement of the electrical conductance of a standard KCl solution at the same temperature according to the equation :

$K = \frac{L}{C}$ in which L=known specific electrical conductance of the standard solution, usually 0.02M KCl in millimhos/cm (1millimho= 1/1000 mho) and C=conductance of the standard solution measured in the given cell (mmhos/cm).

The specific conductance L, of 0.02M KCl is 2.39 mmhos/cm at 18°C, and 2.77 mmhos/cm at 25°C. The measured conductance C, of a test solution in mmhos/cm multiplied by the cell constant (K) gives the specific conductance, L in mmhos/cm of the test solution (L=KC).

The instrument used for this purpose is known as conductivity bridge. A typical system consists of an alternating current (A.C.) Wheatstone bridge, a primary element of conductivity cell and a **null** balance indicator (as in "Solubridge") or an electronic "eye" in the conductivity meter.

Care and maintenance : The conductivity meter or the solubridge (or salt-bridge) has a long life and it rarely goes wrong. If it does, the metallic cover of the meter may be unscrewed and the internal wiring examined for any loose contact, or the vacuum tube may be checked. Quite often, the trouble is found in the conductivity cell. The essential component of the cell is the two electrodes coated with platinum black, and rigidly set at a specific distance (5 mm or so). Sometimes, due to careless handling or inadequate washing, a film of clay is deposited on these electrodes. It can be removed by repeated washings with distilled water or two or three washings with dilute nitric acid (1+4) followed by distilled water. In case the cell needs drastic cleaning then freshly prepared chromic sulphuric acid which is always quite warm is used and the cell finally washed several times with distilled water. The chromic acid must not be allowed to get in contact with the rubber bulb of the conductivity cell or any metallic parts.

Various types of Conductivity Meters

Salt bridge or electrical resistance bridge is manufactured by Wood and Metal Products Co., Bloomfield Hills, Michigan, U.S.A. and Instrument Development and Manufacturing Corp., East Pasadena, California, U.S.A. Industrial Instruments Inc., Jersey City, N.J., U.S.A. and Mullard Equipment Ltd., Mullard House, Torrington Place, London, England are suppliers of conductance bridge suitable for measurement of soluble salt content of soil solution as well as gypsum block resistance (Jackson 1973).

Cambridge conductivity meter (bridge) is a mains (A.C.) operated Wheatstone bridge; there is a built-in 1000 cycles per second oscillator. This instrument is supplied by Cambridge Instrument Co. Ltd., Grosvenor Place, London, U.K. A high precision, conductance bridge is marketed by W.G. Pye and Co. Ltd., New Market Road, Cambridge, England. Messers ELICO (India) Private Ltd. have also developed a conductivity bridge (50 c/s to 1000 c/s) which has a similar type of 'magic eye' detector as in the case of *Solubridge*, soil test model RD-26 of Industrial

Instruments Inc. of U.S.A. Recently, M/s Systronics and some other manufacturers in the country have come out with similar products which, however, need to be tested for their performance.

3. *COLORIMETRY*

Colorimetric analysis is based on the measurement of intensity of radiant energy after it passes through a (coloured) solution. The procedure consists of passing a monochromatic light beam of known intensity through the sample solution and determining the intensity of the transmitted beam with the help of a photo-electric cell. In other words, colorimetry can be considered as *absorption spectrophotometry in the visible range* of light.

The quantitative law basic to colorimetric analysis is Beer's law which may be stated as follows :

$$\log \frac{I_0}{I_t} = k.C. \qquad \ldots\ldots(1)$$

where I_0 = intensity of incident radiation

I_t = intensity of transmitted radiation

C = concentration in solution

k = a constant

The term (1) is the absorbance or optical density of the sample.

It must be realised that the above relationship holds good only where absorption of the incident light occurs in the sample *i.e.*, when the light is of the wavelength at which the sample absorbs. Thus, monochromation (to a very narrow wave-band) of the incident light through suitable filter or grating is an essential requisite of colorimetry/absorption spectrophotometry (Willard *et al*, 1965).

Types of Colorimeter

Instruments for absorption photometry may be classified as visual comparators, filter photometers and spectrophotometers. In terms of construction one also recognises the difference between single beam and double beam light paths and whether the particular (photo-electric) instrument is direct reading or employs balanced circuit. Special features include double mono-chromation and dual wavelength mono-chromators,

plus arrangement for automatic recording (Jackson 1973). In the selection of an instrument, one should consider the initial cost, maintenance, flexibility of operation and adaptation to special situations.

The simplest type of colour comparators work on side-by-side viewing of light from a common source, through a pair of matching flat bottom tubes, one containing the unknown solution and the other the standard for comparison. On matching the two, the ratio of the height (vertical depth) of the solutions would be inversely related to the ratio of their intensity. The Duboscq colorimeter was probably the most widely used colour comparator before the advent of photo-electric instruments (Willard *et al.* 1965).

In a *single beam* direct reading photo-electric colorimeter, the optical path is simply from the light source, through the filter and sample holder, and finally to the detector. EEL colorimeter, Hilgard Biochem, Evelyn photoelectric colorimeter, Lumetron colorimeter model 450 are the various instruments of this type. Several indigenous make (AIMIL, ELICO, Systronics) instruments are also available.

Double beam photo-electric colorimeters employ barrier-layer matching photo cells ; colorimeters of this type are Fisher electro-photometer, Klett-Summerson photo-electric colorimeter and Lumetron photo-electric colorimeter model 402. The incident filtered light is divided into two parallel beams, one passing through the solution in the cuvette (tube) before falling on the measuring photocell and the other part passing directly on to the reference photocell through an adjustable slit, operated by a control knob.

A typical colorimeter will have (i) a light source, usually a tungsten lamp, (ii) monochromator (filter, prism or diffraction grating), (iii) absorption cell, and (iv) photometric system for measurement of intensity of the light beam in terms of electric current generated.

Analytical Procedure

Since Beer's law (more precisely the Beer-Lambert's law as it is employed in colorimetry) relates the concentration to the logarithm of the ratio of incident intensity to the transmitted one, the absolute intensities need not be determined.

The photocurrent from the photo-electric cell is adjusted to a convenient reading (Go) on the measuring galvanometer with the blank

solution in the absorption cell. The blank is replaced by the test solution and the galvanometer reading is taken (Gt). Since the photoelectric cell is chosen to give a linear response to light intensity, therefore

$$\frac{G_0}{G_t} = \frac{I_0}{I_t}$$

In actual practice, the light with the blank solution is adjusted to give 100% reading on a *transmittance scale* or zero on an *absorbance scale*. The readings are continued with a series of standard solutions and then with the test solutions. A calibration curve is drawn relating the concentration of the standards to the absorbance values, using the relationship,

$$\text{Per cent T} = \frac{I_t}{I_0} \times 100$$

where, T is the transmittance. Thus,

$$\log (\% \text{ T}) = \log 100 + \log \frac{I_t}{I_0}$$

$$= 2 - \log \frac{I_0}{I_t} \text{ ; and}$$

the concentrations of the test solutions are obtained from the corresponding absorbance values.

Care and maintenance : No attempt should ever be made to *touch* the galvanometer assembly of the photoelectric colorimeter. If accidently damaged by mechanical shock or ingress of corrosive liquids, the instrument should be examined by a competent person or sent to an authorised workshop. When the electric bulb (usually 100 watt) fails to light and if there is a dangling wire, the bulb has been damaged (fused) beyond repair and needs replacement. If the bulb filament is intact, any loose connections are to be set right. Care must be taken while replacing the projection bulb with one of the *same wattage* (100 W) and of the *same dimensions*. If photo-cells fitted in the instrument lose their sensitivity for any reason, they may be surface cleaned or replaced as may be necessary but *never to be opened*. It is always advisable to use an automatic voltage stabilizer in the AC supply line. An important part of the instrument is the test tube (absorption cell) for holding the coloured solution to be measured. When filled with a liquid the test tube acts as a *cylindrical lens*, converging the light to a sharp line on the photo-electric cell.

If it is not exactly circular in cross section, its *focal length will change at different angles* and so also will the intensity of the light falling on the photo-cell. A *perfectly circular* tube will not *show any deflection* of the galvanometer needle, at any angle of rotation when filled with distilled water. It is also equally important that the walls of the tube are absolutely clear and free from scratches and no liquid is sticking to the outer side while placing it in the socket provided in the colorimeter.

4. *FLAME PHOTOMETRY*

When a substance is heated, it emits radiant energy. The emission becomes stronger with greater excitation of the molecules/atoms. This energy (electromagnetic radiation) is composed of radiations of different wavelengths ; the assemblage of radiation is the *emission spectrum* of the substance. There are three kinds of emission spectra :

(i) *Continuous spectrum*, given out by incandescent solids, consisting of continuous wavelength range, where individual lines are absent ;

(ii) *Band spectrum*, emitted by excited molecules, consisting of individual bands which are actually composed of groups of lines very close to one another ; and

(iii) *Line spectrum*, originating from excited atoms or atomic ions (which exclude polyatomic ions or radicals). These spectra consist of distinct and often widely spaced lines.

Analytical flame photometry is based on the measurement of the intensity of the characteristic line emission of the element to be determined (Jackson 1973). When a solution of a salt is sprayed into a flame (acetylene, propane or liquefied petroleum gas) the salt gets separated into its component atoms because of the high temperature. The energy provided by the flame excites the atoms to higher energy levels. Actually, the orbital electrons are shifted to higher planes from their normal orientation. When the electrons return back to the ground or unexcited state, they emit their *characteristic radiation*. Since the excitation can be to different levels, light (electromagnetic radiation) of several wavelengths can be emitted. However, the intensity of the wavelength corresponding to the most probable transition will be the highest. For each element, such characteristic lines have already been very well identified.

Each individual atom emits one quantum of radiation. Therefore, the intensity of radiation emitting from the flame (which must be otherwise steady and non-luminous) will be proportional to the number of atoms in the flame, that is, to the concentration of the particular element in the flame. This concentration is, in turn, directly related to the content of the element in the test solution.

The instrumental set-up for flame photometric analysis consists of three parts :

(a) *Atomization assembly*, comprising a nebulizer (atomizer) to produce a fine mist from the solution and a burner which vaporizes the analyte (in atomized form) ;

(b) *Monochromation system* (filter, prism) that separates out the analytical wavelength from the other radiations ; and

(c) *Photometric system* for measuring the intensity of the emitted radiation.

Analytical Procedure

A series of standard solutions are prepared and the intensity of emission determined for each concentration after zero setting of the blank. A calibration curve is drawn from these data.

The intensity of emission from the test solution(s) is measured simultaneously and the concentration of the element is read from the calibration curve. Only the linear or near linear portion of the curve should be made use of.

Various types of Flame Photometer

Flame photometers or flame spectrometers are essentially of two basic designs which can be designated as single or double-beam arrangement.

Single-beam instrument : A single beam equipment referred to as direct reading type, comprises only one set of optics. Light emitted from the core of the flame just above the inner cone is collected by a reflector and focussed by a lens of heat-resistant glass through interchangeable optical filters (interference type filter) on to a single photo-detector. Alternatively, light from the burner passes into a monochromator and

radiation leaving the exit slit is focussed on to the photo-detector unit (Jackson 1973).

Double-beam equipment : In the double-beam type instrument, a second light path is provided for the radiation emitted by the internal standard element (*e.g.*, lithium) that is added in a fixed amount to each test solution as well as calibration standard. The wave lengths of internal standard and analytical materials are isolated by means of special filters in a dual optical system and each beam is focussed on separate photoelectric cell (Willard *et al.* 1965).

The various types of flame photometers available commercially are classified into four main categories *viz.*, (i) direct-reading, absorption filters, barrier-layer cell (Coleman, Evans Electroselenium, EEL-Corning models 2 and 3, Schukhnecht, Systronics, ELICO); (ii) internal stndard, interference filters, barrier-layer cell (Analytical Instruments, Baird model KY-2, Electrosynthese, Janke, Lange model 4, Norelco, Patwin, Barclay) ; (iii) prism mono-chromator, direct reading or recording, photo-multiplier tube (Beckman model B-DB, Beckman model DU, Unicam SP 900, Hilger UV Spek, Perkin-Elmer model 52C) ; and (iv) grating monochromator, recording photomultiplier tube (Jerrell-Ash, Ebert, Hitachi, Perkin-Elmer model 139).

Care and maintenance : Although a very sensitive instrument, flame photometers are usually fairly sturdy and give trouble-free service for years if handled properly. The burner assembly has two external connections, one leading in gas directly to the burner and the other taking compressed air through a spraying nozzle (atomizer). The air stream carries with it to the burner flame a spray of the solution under test. If any chocking or disconnection takes place it should be located and can be set right easily. The gas and air pressure *must remain steady* during the operation of the instrument and the connections and regulators should be checked before use. It is of utmost importance that only very clear solutions/extracts are fed in to avoid any possible chocking of the capillary. Burners of flame photometers should be cleaned periodically. While closing down, the gas supply should be first turned off and then the compressed air line. When large salt concentrations or strong acid solutions are to be fed in (which should be *avoided* as far as possible) distilled water should be run for some time and the burner cleaned *immediately* after use. The galvanometer assembly must not be touched and any repairs should be carried out at a competent workshop.

5. *ATOMIC ABSORPTION SPECTROPHOTOMETRY*

The principle behind analytical atomic absorption spectrophotometry (AAS) is the measurement of the absorption of light (radiant) energy by free atoms in the ground state, and its quantitative correlation with the concentration of the analyte in solution.

Absorption of light by atoms : When a solution of a salt is aspirated into a flame the solute molecules dissociate into neutral atoms (fundamental, ground or unexcited state) and excited neutral atoms and ions. In the present day atomic absorption spectrometry one is concerned with the fundamental state atoms only. The number of fundamental state atoms in the flame (acetylene, propane, LPG) is far in excess of that of excited ones. (*Excited atoms* are responsible for emission of radiations which are measured in analytical *flame photometry*, as described earlier under item 4).

When light beam of the appropriate wavelength is passed through the cloud of atoms in the flame the incident radiation is subjected to absorption by the neutral fundamental state atoms. Obviously, these atoms now reach the excited state. The absorption follows the Beer-Lambert law, *i.e.*, the absorbance is proportional to the number of ground state atoms (Willard *et al.* 1965). It must be noted that this relationship holds good only for the specific (resonance) wavelength (similar to absorption spectrophotometry of solutions where it is of course, molecular absorption).

The atoms of the salt solution excited by absorption of the incident radiation, emit radiation when they return back to the ground state. Since the wavelength of this radiation is not the same as that of the incident light beam (absorbed radiation), this emission will *interfere* with absorbance measurement. In AAS this difficulty is obviated by employing a light source of a specific pulse frequency and matching it with an AC amplifier which will respond to signals of that particular frequency only (Willard *et al.* 1965). Thus, the emitted radiation (which is a continuous one) will not be measured by the photometric system.

Instrumentation

The component systems of an AAS unit are :

(i) Light source

(ii) Atomization assembly (including burner)

(iii) Monochromator, and

(iv) Detector and readout assembly.

Light source : The lamp used should emit the resonance radiation of the element the content of which is to be determined. It should be sufficiently intense and stable to permit reliable measurement and must have a low background emission (*low intensity of interfering lines*).

Hollow cathode lamps with the element to be analyzed as the cathode and filled with an inert gas at low pressure, usually constitute the light source. These give the characteristic emission spectrum of the cathode material. This spectrum contains the resonance wavelength radiation along with other light radiations.

Atomization system : The atomization assembly consists of a nebulizer to produce uniformly fine fog of droplets from the solution to be analyzed, and a burner. The unit is more or less the same as that used in flame photometry, except that the burner has a long and narrow slot at the top (like a fish-tail burner) so that the flame which is analogous to the *absorption cell* in colorimetry, provides a long absorption path for the incident radiation. The fuel-oxidant system used may be acetylene-air, acetylene-oxygen, acetylene-nitrous oxide, hydrogen-air, hydrogen-oxygen, propane-air, propane-oxygen or LPG-oxygen/air.

Monochromator : The monochromator enables the isolation of the resonance wavelength from other radiations before measurement of intensity. A diffraction grating or prism is normally used. Filters are usually not adequate for complete monochromation.

Detection and readout system : This unit consists of a photo-sensitive component (photo-multipliers are commonly employed), an AC amplifier, and a sensitive galvanometer to measure the current.

Analytical Procedure

The absorbance values corresponding to known standard solutions are recorded after necessary blank setting and a calibration curve prepared. The absorbance of the test solution is determined and the concentration read off from the calibration curve.

The calibration *must be done for each set of analysis* since in practice it is not possible to reproduce *exactly* the nebulization and flame conditions.

AAS technique of analysis of non-metals is not easily done since their resonance lines mostly occur in the vacuum ultraviolet region. However, some indirect methods can be developed based on the effect of anions on the absorption by metals but these may not be as sensitive. For some elements special types of burners and lamps have been developed with variable degree of success.

Types of Atomic Absorption Spectrophotometer

Some of the established manufacturers of the different types and models of atomic absorption spectrophotometers are :

(1) Atomic Absorption and Electronics Corporation, New York, (2) Aztec Instruments Inc., Westport, Conn., (3) Bausch & Laumb Inc., Rochester, New York, (4) Beckman Instruments Inc., Fullerton, Calif., all from U.S A., (5) Evans Electro Selenium Ltd., Essex, England, (6) Hewlett-Packard, Dayton, Ohio., (7) Jerrell-Ash Co., Waltham, Mass, both of U. S. A., (8) Hilger and Watts Ltd., London, England, (9) Optica S. P. A., Milano, Italy, (10) Perkin-Elmer Corporation, Norwalk, Conn., U. S. A., (11) Techtron Pty. Ltd., East Oakleigh, Victoria, Australia, (12) Unicam Instruments Ltd., Cambridge, England, (13) Carl Zeiss, Oberkochen, East Germany, (14) Jobin-Yvon Arcueil (Seine), France, and (15) Hitachi Instrument Co., Tokyo, Japan.

Important points regarding operation and maintenance of the atomic absorption spectrophotometer : Being one of the quite sophisticated, high precision and fairly expensive type of measuring instrument, the AAS, requires *very careful handling and maintenance.*

The light source is a *critical* component. There should not be any erratic fluctuations in the intensity. Any fluctuations in the line voltage must be taken care of. Lamps are generally operated at low currents to improve linearity of response and maintain narrow emission lines, as much as possible.

While extinguishing the flame the fuel supply should be *first turned off* before stopping the oxygen or air line. The oxygen (or air) and fuel pressure *must be adjusted* to the value recommended in the instruction manual furnished by the manufacturer. The width of the slit is usually set at 0.02-0.05 mm as instructed in the working manual. The wavelength should also be set properly *for each determination* by making use of the relevant components like filter photometer monochromator.

The various aspects to be considered for optimum working conditions are : the emission intensity, steady rate of atomization, the resolution of the spectral lines (or bands) from each other and from adjacent spectral features of other sample constituents, the magnitude of the background radiation, and steadiness of readings.

The AAS can give reliable and trouble-free service only with *careful handling*. Erratic readings may come from fluctuations in gas and air pressure and electric supply or the lamps becoming weak due to prolonged use. If something goes wrong, the instrument should *never be opened* except by a qualified mechanic/technician who should preferably be acredited by the manufacturers or their authorised agents for carrying out the repairing job.

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Appendix I

Division of Soil Science and Agricultural Chemistry

Indian Agricultural Research Institute

NEW DELHI - 110012

INFORMATION SHEET TO ACCOMPANY SOIL SAMPLES

Name of farmer/owner_____ Address_____

Village_____ P O._____ District_____ State_____

Serial No.	Laboratory No. (To be given at the ST Laby)	Sample No. or identification mark	Depth of sampling	Number of samples mixed to get representative sample	**Area** (in acres/ hectares)	**Slope** (Level/ undulating)
1	2	3	4	5	6	7

...contd next page

Appendix I (contd)

Elevation	Drainage	Irrigation	Source of irrigation	Salinity	Crop growth	Recommendations
Upland/ lowland	Good/poor	Irrigated/ rainfed	If irrigated	Saline/ alkali or not	Good/poor/ average	(Sought f which cro
8	9	10	11	12	13	14

Any other relevant information to be furnished

Place :
Date :

To
SOIL TESTING LABORATORY
(Location and address)

Signature

Appendix II

RATING CHART FOR SOIL TEST DATA

Nutrient	Fertility status rating*		
	Low	Medium	High
Organic carbon : as a measure of available N (%)	Below 0.50	0.50-0 75	Above 0.75
Available P (kg/ha)	Below 11	11-25	Above 25
Available K (kg/ha)	Below 120	120-280	Above 280

*Subject to marginal adjustment as warranted by local conditions

Appendix III

FERTILIZER RECOMMENDATION SCHEDULE ON SOIL TEST BASIS

(For the union territory of Delhi and neighbouring region)

The recommendations giving the dose of fertilizer nutrients (NPK) to be supplied according to the soil test values are meant for **high yielding varieties** of wheat, rice and maize grown under **assured water supply** and with an yield expectancy of **40 to 45 q/ha.** These may be largely applicable to north Indian (alluvial) plains having more or less similar type of soil and climatic features.

N.B. This schedule can also serve as a model. Necessary adjustments in fertilizer doses are to be made according to crops and their yield levels, soil, climate and other local conditions.

NITROGEN		PHOSPHORUS		POTASSIUM	
Soil test value (per cent org. C)	Kg N/ha as fertilizer to be applied	Soil test value (Olsen's P : kg/ha)	Kg P_2O_5/ha as fertilizer to be applied	Soil test value (amm. acetate—K : kg/ha)	Kg K_2O/ha as fertilizer to be applied
Up to 0.20**	120**	Up to 10	80	Up to 100	60
0.21- 0.40	120*	11-20	60	101-150	50
0.41—0.60	120	21-30	45	151-200	40
0.61 - 0.80	90	31-40	30	201-250	30
0.81—1.00	75	41-55	20	251-300	20
Above 1.00	60	Above 55	10	Above 300	0

**In soils with such low content of organic matter application of farmyard manure @ 20 tonnes per hectare must be made in the *kharif* (monsoon) season at least one month before sowing.

*To be supplemented with FYM dose of 10 t/ha

Appendix IV

INDIAN AGRICULTURAL RESEARCH INSTITUTE, NEW DELHI

DIVISION OF SOIL SCIENCE AND AGRICULTURAL CHEMISTRY

(Soil Testing Laboratory)

Soil test report and recommendation sheet

Name : Village/Taluk . P. O : District : State :

Sl. No.	Lab. No	Sample mark	pH (soil reaction)	E. C.* (m mhos/cm)	Org. carbon (%) **	Av. phosphorus (lb/ac) (× 1.12 = kg P/ha)	Av. potash (lb/ac) (× 1.12 = kg K/ha)	Calcium carbonate ***	Texture ***	Crops	Fertilizer recommendation: FYM (ton/ac.)	CAN (kg/ac.)	Urea (kg/ac.)	Single super-phosphate (kg/ac.)	DAP (kg/ac.)	Muriate of potash (kg/ac.)	Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

* Measure of total soluble salts

** Measure of available soil nitrogen

*** Approximate values by rapid determination

The recommendations are for the high yielding varieties of crops (wheat, rice, maize etc.) For horticultural crops and flowering plants the concerned Specialist or State Deptt may be consulted for specific recommendations.

L : Low
M Medium
H : High

Scientist

Appendix V

SALINITY APPRAISAL OF SOIL FROM ELECTRICAL CONDUCTIVITY (EC) MEASUREMENT (1 : 2 SOIL - WATER EXTRACT)

Soil texture	E.C. in mmhos/cm at 25°C		
	Limit for		
	sensitive crops	semi-tolerant crops	tolerant crops
Clay	1.2	2.4	4.8
Clay loam	0.8	1.6	3.2
Loam	0.7	1.4	2.8
Sandy clay loam	0.6	1.2	2.4
Sandy loam	0.4	0.8	1.6

Appendix VI

ELECTRICAL CONDUCTIVITY AND SALINITY CHARACTERISTICS AND THEIR RELATIONSHIPS

Soluble salts : In milliequivalents per litre (m.e./l) = EC in mmhos/cm × 10 (*approx.*)

Conductivity and osmotic pressure

O.P. (atm.) = 0.36 × EC in mmhos/cm for soil extract in the range 3 to 30 mmhos/cm at 25° C

Conductivity to ppm : ppm = 0.64 × EC × 10^6 for irrigation water in the range 100 to 5000 micromhos/cm
(EC in mhos × 10^3 = millimhos and × 10^6 = micromhos)

Gypsum in tons per acre foot of water (taf) = 0.000136 × ppm of calcium

Saturated gypsum solution contains at 25°C 30.5 m e./1, 2.63 g/1, 2630 ppm and 3.5 tons of gypsum per acre foot cf water.

Appendix VII

SALT TOLERANCE OF CROPS (in descending order)

Sensitive	
Field crops	Field beans, green gram
Fodders (forage)	Red clover, white clover, *guar* (*Cyamopsis* sp.)
Vegetables	Celery, radish (English var.), green bean
Fruits	Peach, apricot, pear, apple, plum, strawberry, blackberry, pineapple
Semi-tolerant	
Field crops	Oats, rice, *jowar* (sorghum), maize, *bajra* (pearl millet), wheat (improved var.), *arhar* (pigeon pea), castor, flax, jute, soybean, rye, gram, castorbean
Fodders (forage)	Lucerne, *senji* (*Metilotus* sp.), mentha, *jowar*, maize, berseem, cowpea, ryegrass, sweet clover, Sudan grass, oat grass
Vegetables	Tomato, cabbage, cauliflower, lettuce, potato, radish, carrot, onion, pea, cucumber, pumpkin, sweet potato, squash
Fruits	Grape, olive, fig, guava, mango, banana, pomegranate, orange, grapefruit, lemon, almond
Salt tolerant	
Field crops	Barley, *dhaincha* (*Sesbania* sp.), sugarbeet, tobacco, cotton, wheat (local var.), sugarcane, *taramira* (*Brassica compestris*), rapeseed
Fodders (forage)	Salt grass, *dub* grass (*Cyanodon* sp.), Rhodes grass, Bermuda grass
Vegetables	Beetroot, asparagus, spinach, kale, turnip
Fruits	Date palm, coconut, *falsa* (*Grewia asiatica*)

Appendix VIII

NORMALITY, SPECIFIC GRAVITY AND PERCENT BY WEIGHT OF SOME COMMON REAGENTS IN THE LABORATORY

Reagent	Normality	Specific gravity (20°C)	Per cent by weight	One ml contains (g)
Acetic acid (glacial)	17.5	1.06	99 $H_4C_2O_2$	1.06 $H_4C_2O_2$
Ammonium hydroxide (conc.)	15.0	0.88	28 NH_3	0.26 NH_3
Hydrochloric acid (conc.)	11.8	1.19	39 HCl	0.47 HCl
Nitric acid (conc.)	15.6	1.42	71 HNO_3	0.99 HNO_3
Phosphoric acid (conc.)	15.0	1.70	85 H_3PO_4	1.45 H_3PO_4
Sulphuric acid (conc.)	36.0	1.84	96 H_2SO_4	1.76 H_2SO_4
Ethyl alcohol (ethanol)	—	0.79	—	0.79 C_2H_5OH

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Nitric acid (conc.)	15.6	1.42	71 HNO_3	0.99 HNO_3
Phosphoric acid (conc.)	15.0	1.70	85 H_3PO_4	1.45 H_3PO_4
Sulphuric acid (conc.)	36.0	1.84	96 H_2SO_4	1.76 H_2SO_4
Ethyl alcohol (ethanol)	—	0.79	—	0.79 C_2H_5OH

Appendix IX

SOME COMMONLY USED PRIMARY STANDARD SOLUTIONS

Reagent	Amount of AR grade chemical dissolved and made to one litre to give 0.1N solution
Potassium hydrogen phthalate	20.422 g
Oxalic acid,crystalline dihydrate	3.165 g
Sodium carbonate, dehydrated at 260—70°C	5.204 g
Potassium dichromate	4.904 g
Sodium chloride	5.846 g
Borax($Na_2B_4O_7.10H_2O$)	19.070 g

Appendix X

SOME CONVERSION FACTORS

Per cent P_2O_5	× 0.44	=	Per cent P
Per cent P	× 2.29	=	Per cent P_2O_5
Per cent K_2O	× 0.83	=	Per cent K
Per cent K	× 1.20	=	Per cent K_2O
Per cent Ca	× 1.40	=	Per cent CaO
Per cent Mg	× 1.67	=	Per cent MgO

ppm = milliequivalent/litre × equiv. wt
= microgram per ml or mg per litre

m.e. per litre = ppm ÷ equivalent wt
1 acre = 4840 sq. yd = 4047 sq. m = 43560 sq. ft = 0.404686 hectare
1 hectare = 2.471 acre
lb/acre × 1.121 = kg/ha
One lb = 0.4536 kg : 1 kg = 2.2046 lb

ANNEXURE I

FERTILIZER RECOMMENDATIONS ON SOIL TEST BASIS—VARIOUS APPROACHES

Several approaches on fertilizer recommendations based on soil test are being tried in India so as to maximise the production of crops per unit of fertilizer input. One such is based on the results of **agronomic experiments** on farmers' fields and categorizing the soil (arbitrarily) into **low, medium** and **high** in respect of status of available soil nutrients. This is being currently followed by a large number of soil testing laboratories in the country. In some cases further sub-classifications *e.g.*, very low, low, medium, high, very high etc. have been made by considering narrow ranges in the soil test values as given in appendix III. The second one known as the **critical level** concept, advanced by Cate and Nelson[1], is the status of available nutrient in the plough layer of soil below which an economic response is expected from the application of that particular nutrient and above which the probability of such response is low. Although experimental data and follow up work are rather limited, this approach has also proved quite useful (Goswami *et al.*[2]).

The other two types of procedure envisage the evolving of soil and fertilizer nutrient adjustment equations *viz.*, (i) **multiple regression analysis** of available soil nutrients (soil test values), fertilizer doses and yield of crops for economic and optimum production and (ii) prescription procedure or the **targetted yield concept** (Ramamoorthy *et al.*[3]) wherein fertilizer doses for a specific yield target are calculated from

1. Cate, R.B. Jr. and Nelson, L.A. A simple statistical procedure for partitioning soil test correlation data into two classes. *Proc. Soil Sci. Soc. Am.* **35** : 658-60; 1971
2. Goswami, N.N., Bapat, S.R. and Pathak, V.N. Studies on the relationship between soil tests and crop responses to phosphorus under field conditions. *Proc. int. Symp. Soil Fert. Evaln*, **1** : 351-59; 1971
3. Ramamoorthy, B., Narasimhan, R.L. and Dinesh, R.S. Fertilizer application for specific yield targets of Sonora-64 wheat. *Indian Frng*, **17** (5) : 43-45; 1967

the nutrient requirement per quintal of grain production of a particular crop variety, percentage contribution of nutrients from the soil and that of fertilizers. Various other approaches for soil fertility evaluation and optimum fertilizer inputs have been discussed by Goswami and Biswas[4].

A fairly large number of replicated experiments have been conducted on farmers' fields in the Union Territory of Delhi to evaluate these procedures and a few examples of the results obtained with wheat crop are cited in tables 1 and 2 (Bajaj[5]).

Fertilizer recommendations on soil test basis have proved definitely more profitable than *ad-hoc* application of standard doses (according to the general package of practices). The response ratio was still higher where **adjustment equation** for economic yield by multiple regression technique was adopted (Table 1).

Table 1 : *Some results of fertilizer recommendations based on soil tests for wheat* (var. Sonalika)

Location (village)	Soil test values (kg/ha)			Fertilizer doses (kg/ha)			Yield (q/ha)	Fertilizer response ratio
	N	P	K	N	P_2O_5	K_2O		
(i) Procedure adopted in the ST Laby, IARI (as in appendix III)								
Goela	0.62*	12.0	210	90	60	30	37.5	12.3
Sanoth	0.86*	6.5	220	75	80	30	38.0	11.8
Ghoga (1)	0.82*	15.0	180	75	60	40	38.5	11.1
Ghoga (2)	0.45*	37.0	350	120	30	0	36.0	14.0
					Average :		37.5	12.3
(ii) Multiple regression technique								
Bawana	156**	4.5	200	220	91	0	57.4	10.2
Jharoda (1)	177**	11.3	497	185	91	0	57.4	10.8
Jharoda (2)	200**	7.2	300	143	91	0	52.0	15.3
Tajpur	180**	4.5	154	180	91	20	53.8	14.8
Bharthal	202**	17.3	320	147	73	0	55.7	13.2
					Average :		55.2	12.9

*Per cent organic carbon taken as the measure of available soil nitrogen

**Available N(kg/ha)

4. Goswami, N.N. and Biswas, T.D. Soil testing for fertilizer recommendations. *Fertil. News*, 19(8) : 24-37; 1974

5. Bajaj, J.C. Evaluation of various methods of making fertilizer recommendations for cereal crops in the cultivators' fields. *Trans. 12th int. Congr. Soil Sci. New Delhi*, 6 :p. 110 (Abstract); 1982